DBA-VC

Vendor Companion to the DBA Series

AI and Command Broker Requirements

Cross-Series Instrument

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Audience: ICS/OT Vendors, Suppliers, and System Integrators — Product Management, Engineering, and Business Development

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1. Purpose and Scope

This document is addressed to the vendors, suppliers, and system integrators who build and deliver industrial control systems (ICS) and operational technology (OT) products to critical infrastructure asset owners and operators. Its purpose is direct: to translate the AI and Command Broker requirements embedded in the DBA series into product specifications that a product manager, an engineering lead, or a business development executive can act on.

The DBA series — the Design Basis series published by Eclectic Technologies — extends the Design Basis Accident methodology of nuclear safety engineering across critical infrastructure sectors. The series now carries a sector framework for each of the sixteen critical infrastructure sectors. Energy is carried by DBA-EN-SF (*Energy Sector Framework*) and its three sub-sector frameworks, DBA-ES (Electric), DBA-GAS (Natural Gas) and DBA-OIL (Petroleum), with data centers carried by DBA-DC within the Electric sub-sector's Critical Loads grouping; nuclear by DBA-NUC-SF (*Nuclear Sector Framework*); water by DBA-W; and the remaining thirteen sectors by their own frameworks, from chemical (DBA-CHEM) through transportation (DBA-TRAN). The military-action extension is carried by DBA-MA. Each sector framework carries requirements for the deployment of machine learning (ML) and generative AI (Gen AI) in OT control systems. Those requirements are stated in sector-specific operational vocabulary. This document abstracts them into canonical, sector-neutral product specifications and maps each specification to the sector frameworks in which it appears.

This document is a companion to the entire DBA series — not to any single sector. The Bright Line and Command Broker architecture established in Part I applies wherever a vendor's product operates, regardless of the sector label on the facility. A vendor supplying an ICS platform that is deployed across energy, water, nuclear, and transportation facilities carries the same cross-sector obligation at every installation. Part I establishes that obligation in sector-neutral terms. Part II instantiates it for the two most mature DBA sectors — energy and nuclear — and provides framework stubs for sectors where DBA work is in progress.

1.1 What This Document Is Not

This document does not establish a certification scheme. No DBA-series certification program currently exists. A vendor who builds to the specifications in Part I cannot yet receive a compliance certification, but can assemble the evidentiary package described in §6 that will support such a claim when procurement language begins to reference DBA-series requirements explicitly.

This document does not replace the sector framework documents. Vendors serving specific sectors should read the applicable sector framework for the full operational context of these requirements. DBA-VC provides the abstracted, cross-sector product specification; the sector frameworks provide the operational context in which the product will be deployed and assessed.

PART I — CROSS-SECTOR FOUNDATION

2. Analytical Foundation

The DBA series AI and Command Broker requirements derive from two analytical sources. The first is the risk-landscape study Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers (Roxey, T.E., Version 29, Eclectic Technologies, 2026). The second is the Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector), published by Eclectic Technologies, and its implementing instruments — the CB-IB (Command Broker / Influence Broker) Qualification and Onboarding Standard, the Command Broker Implementation Guide, and the Command Broker use-case series (CB-UC-1). This section summarizes the key distinctions that govern product design requirements.

Governing definitions. This document specifies vendor product requirements. It does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application, and the obligation it creates — is defined in FD-BL (Foundational Definition — The Bright Line), which governs. The Broker role that discharges that obligation, the Line-to-Broker pairing invariant, the competence rule, and the conditions under which a verified gatekeeper may stand in for the Broker's presence are defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary, and the machine-learning carve-out are defined in FD-LD. Two determinations bind under FD-BL: FD-BL-D1 governs the mode by which the authorization obligation may be discharged, and FD-BL-D2 governs the integrity of the basis on which an authorization is granted. FD-EV reserves the term "envelope." Where this document and any of those instruments differ, the Foundational Definition or determination governs, and this document propagates by reference rather than by restatement (FD-BL §2.1).

2.1 Scope — The Determinism Carve-Out

Scope and placement. These are two distinct questions, and this section answers only the first. Whether a component is subject to Bright Line analysis at all is answered by the determinism carve-out, owned by FD-LD §4.3, which FD-BL §4.4 cites: a component deterministic within a stated input bound and formally testable as deployed is outside the analysis; every component that does not meet the carve-out is within it, a component whose outputs are not reproducible under identical inputs being the principal case, whatever its architecture is named. Where the line falls for an in-scope component's actions is a separate question, answered on consequence at §2.3. A vendor relying on the carve-out shall state the input bound over which determinism holds and have it verified as part of assessment; an exemption asserted as "merely deterministic," without that bound, is not a valid carve-out (FD-LD §4.3).

Machine learning (ML) encompasses algorithms that learn from data to improve performance — anomaly detection, predictive maintenance, pattern recognition, and load forecasting. Such

algorithms are frequently deterministic: given inputs within a stated bound, they produce bounded, testable outputs verifiable through semi-formal methods. An anomaly detection algorithm running on a SCADA historian is a common example. Where its behavior is deterministic within a stated, verified input bound and formally testable as deployed, it is outside Bright Line analysis under the carve-out — which is a scope finding, not a permission, and not a finding that its actions are below the line (FD-LD §4.3).

Generative AI (Gen AI) encompasses large language models and similar systems that produce novel outputs from learned distributions. Gen AI is non-deterministic: the same input can produce different outputs across execution cycles, and formal verification methods do not apply. A large language model assistant proposed for HMI integration or operational decision support is non-deterministic and is therefore within scope of Bright Line analysis. Being in scope does not place any of its actions. Each action it produces is placed on its own consequence under §2.3, and only the actions above the line require the authorization at §3.1.

The vendor's obligation is to determine, for each component, whether it satisfies the property test stated at the head of this section, and to state the input bound over which determinism holds wherever the carve-out is relied upon. A product combining components of both kinds assesses each independently (FD-LD §4.3). Scope is not decided by product category, and it is not decided by the criticality band of the application.

2.2 The Criticality Bands

The CB-Framework establishes three criticality bands based on the severity of the worst credible consequence, aggregated across the six consequence categories by taking the maximum (Consequence-Severity Tier Model). Band assignment is determined by the consequence of the application, not the capability of the AI component.

Highest-Criticality band. Applications whose malfunction could produce Design Basis-level consequences: population-scale public health impact, loss of critical infrastructure function, or life-safety consequences. The band sets the rigor of assessment, qualification, and audit that applies to the application. It does not decide, for any individual action, whether Command Broker authorization is required: that is the Bright Line placement question at §2.3, answered action by action on consequence (FD-BL §4.3). A Highest-Criticality application contains below-line actions, and the product's obligation is to classify and route each action, not to gate the application.

Mid-Criticality band. Applications involving moderate consequences that do not rise to the Design Basis threshold. A reduced but substantive assessment, qualification, and audit regime applies. As with the Highest band, whether any specific action may proceed without authorization is decided by its placement at §2.3, not by the band and not by whether the producing component is ML or Gen AI.

Lowest-Criticality band. Applications whose consequences do not affect critical functions or life-safety in any category. The lightest assessment and audit regime applies. A Lowest-Criticality application may nonetheless contain above-line actions, and where it does, those actions carry the authorization obligation in full (FD-BL §4.3).

The band classification of an application is the joint responsibility of the vendor and the asset owner/operator (AOO). The vendor must document the band classification for which their product is designed and the configuration changes required to deploy at a different band. The AOO assigns the final band classification for their facility's specific application context. The commissioning gate at which this joint classification is recorded and signed off is specified in CB-CT Commissioning & Joint Training (§4.1).

Naming crosswalk (legacy reading aid). The band names used here are canonical. Earlier documents in the corpus, and the ISA Parent Framework, numbered the same bands Tier 1 / 2 / 3; that numbering is retained below only to read older text. The parenthetical type-words — safety-critical, quality-of-service, administrative — are not band definitions; they name error classes and are defined as such in the Tiered Assessment Framework.

Criticality band (canonical)	Legacy number	Severity — worst credible consequence, any category
Highest	Tier 1	Design-Basis level: population- or mission-scale, loss of a critical function, or life-safety
Mid	Tier 2	Significant but bounded and recoverable, below the Design-Basis threshold
Lowest	Tier 3	No reach to a critical function, life-safety, or critical service in any category

2.3 The Bright Line

The Bright Line and its placement rule are defined in the Foundational Definition — The Bright Line (FD-BL, §4.1–§4.5), which governs. The application-band-versus-action-level distinction stated here is the governing model of FD-BL §4.3, and this section is the source text from which that provision was drawn.

The Bright Line is a consequence threshold, not a technology classification or a fixed layer boundary. It is the line — drawn action by action — between actions whose erroneous execution could produce a Design Basis-level consequence, which require substantive and attributable human authorization, and actions whose erroneous execution produces a correctable, facility-local consequence, which may execute under a lighter confirmation. Nothing above the line reaches physical command execution without that authorization, and the separation is enforced by architecture, not by procedure or operator discipline. Authorization is ordinarily discharged in

the moment by a human Command Broker; for a narrow class of actions it may be discharged in advance, under the conditions at §3.1 (FD-BL-D1).

The Bright Line is not a configuration option, a policy statement, or a vendor assurance. It is an engineering requirement: the control system must be mechanically incapable of executing an above-line action originating from a component in scope under FD-LD §4.3 without the required authorization. Where that authorization is discharged in the moment, the confirmation step must be local, auditable, and not itself executable remotely or by another software component. The enforcement mechanism shall not be defeasible by configuration change, software update, or system integration (FD-BL §4.5).

Note on naming. The term "Bright Line" is used exclusively in the FD-BL sense throughout all current DBA-series instruments. In the DBA-GAS and DBA-OIL sector frameworks, federal coordination trigger language uses "Federal Coordination Threshold," not "Bright Line." These are distinct concepts with distinct terms. The term "envelope," and the compound "pre-authorized envelope," are likewise reserved — for the authorization sense: a region within which an above-line action is authorized in advance, whose completeness proof has been accepted and a change to which is categorically controlled (FD-EV §4.1). The operational sense — a region of normal operation whose violation is a runtime event to detect and report — is the operating band, its boundary an operating limit, and the approach of operation toward that limit encroachment (FD-EV §4.3).

2.4 Cross-Sector ICS Amplifier

Many ICS and OT vendors do not serve a single sector. A SCADA platform, a distributed control system, an HMI suite, or a Gen AI operator decision support tool may be deployed across energy facilities, water treatment plants, nuclear power stations, transportation infrastructure, and data centers — often by the same vendor, sometimes in the same product line. This is the cross-sector amplifier: the Bright Line and Command Broker architecture follows the product, not the sector boundary.

The commercial implication is symmetrical. A vendor who builds to the DBA-series standard — Command Broker enforcement active by default, documented downgrade pathway, adversarial ML robustness specifications — has a compliance posture that applies across every sector in which their product is deployed. The qualification record built for a DBA-EN (energy) deployment is directly transferable to a DBA-W (water) or DBA-NUC-SF (nuclear) deployment. Vendors who build to the standard once do not rebuild sector by sector. Vendors who do not build to the standard carry a compliance deficit that compounds across every sector they serve.

The practical market signal: as DBA-series language propagates through engineering firm practice and utility procurement specifications, it will appear across sectors near-simultaneously. A vendor who has completed the evidentiary package in §6 is positioned to respond. A vendor

who has not must retrofit under procurement deadline pressure, simultaneously, across multiple sectors.

2.5 Vendor Applicability Self-Assessment

Not every vendor product carries the same DBA-series obligation. The obligation scales with the product's proximity to physical command execution and the criticality of the application. This section provides narrative guidance for self-assessment, followed by a one-page checklist for quick triage.

The self-assessment proceeds in four steps. First, identify whether the product contains any AI component — ML or Gen AI — that has any pathway, direct or through integration, to a physical process parameter. If the product has no such pathway, the AI governance requirements in Part I do not apply to it, though supply chain assurance requirements may still apply to embedded software components.

Second, if a pathway exists, determine the criticality band of the application context. A product deployed in a water treatment disinfection system, a pipeline compressor station, a nuclear reactor protection system interface, or an energy management system dispatch function is in a highest-criticality application. The full Part I requirements apply.

Third, determine whether each AI component is within scope. A component deterministic within a stated input bound and formally testable as deployed is outside Bright Line analysis under the carve-out — which is not a finding that its actions are below the line, and not a grant of ungoverned autonomy (FD-LD §4.3). The carve-out holds only for the configuration that was formally tested: an action produced while the component runs any other configuration, including one altered outside a governed act, is not covered by it (FD-LD §4.3; Local Sovereignty §3.2). Every component that does not meet the carve-out is within scope — a non-deterministic component, and equally a deterministic one never formally tested, or one whose input bound is not stated — and each of its actions is then placed on consequence. Products combining both assess each component independently, and any component relying on the carve-out states the input bound over which determinism holds.

Fourth, identify which sector-specific chapters in Part II apply to the product's deployment contexts. A vendor serving only water utilities reads §10.1 and the DBA-W references. A vendor serving energy and nuclear reads §8 (Energy Sector) and §9 (Nuclear Sector). A vendor serving multiple sectors reads all applicable chapters and treats the most stringent requirement as the floor.

Vendor Applicability Checklist

Question	If Yes →
Does your product contain any AI component (ML or Gen AI) with any pathway to a physical process	Continue to the next question. If No: supply chain assurance requirements may still apply to embedded

parameter?	software.
Does your product operate in a highest-criticality application (water treatment, pipeline, nuclear, energy dispatch, or equivalent)?	Full Part I requirements apply. Proceed to §3 through §6.
Does your product operate above the Bright Line in any facility class covered by this series?	Command Broker enforcement is mandatory. See §3.1 and §3.2.
Does your product contain any component that is non-deterministic — outputs not reproducible under identical inputs — or whose determinism is not established over a stated, verified input bound?	The component is within scope. Each of its actions is placed on consequence; above-line actions require authorization under §3.1. Red-team obligation applies. See §3.1, §3.3, §3.6.
Does your product serve multiple sectors (energy, water, nuclear, transportation)?	Cross-sector amplifier applies. The most stringent sector requirement is the floor. See §2.4 and applicable Part II chapters.
Is your product an EMS platform serving Balancing Authority, RC, or TOP functions?	The Energy Sector chapter applies with CntlCo-specific depth. See Part II §8.
Does your product include I&C systems, safety system interfaces, or digital protection systems for nuclear facilities?	The Nuclear Sector chapter applies. See Part II §9.

3. Canonical Product Requirements

This section states each AI and Command Broker product requirement in canonical, sector-neutral form. Each requirement is organized into three obligation levels reflecting the default/downgrade/risk-transfer framework:

- DEFAULT (Must): What the product must do out of the box, without any configuration action by the AOO or integrator. This is the vendor's design obligation.
- **DOWNGRADE PATHWAY (Should):** The conditions under which the compliance obligation may be transferred to the AOO or integrator, and what the vendor must do to make that transfer a documented, risk-visible act. A downgrade transfers the obligation; it does not modify the enforcement mechanism, which is not defeasible by configuration change, software update, or system integration (FD-BL §4.5), and it does not alter the placement of any action.
- RISK TRANSFER OBLIGATION: What the vendor's product documentation must state when a downgrade is executed.

The Default/Should structure is not a weakening of the requirement. It is a recognition that the AOO holds authority and accountability for its facility's design basis compliance. A vendor who

ships with enforcement on by default, and who requires a positive documented configuration action before any transfer of the compliance obligation, has discharged their design obligation.

3.1 Command Broker Enforcement

DEFAULT (Must): The product ships with Command Broker enforcement active. No component in scope under FD-LD §4.3 has a pathway to execution of an above-line action without authorization. Authorization is discharged in one of two modes, and the product documentation shall state which mode applies to each above-line action class.

In the moment (default). A positive human confirmation by a qualified Command Broker, having the properties CB-Framework_Cross_Sector §3.1 requires of it. The product implements those properties and does not vary them.

In advance, by the Command Broker, with a verified gatekeeper admitting each action (FD-BL-D1; The Broker §4.4). Available only for a narrow class of above-line actions, and only where all three conditions hold. **R1:** the whole safety case lives in a deterministic, formally verified gatekeeper that the proposing component can neither modify nor bypass, the proposer holding proposal authority only and the gatekeeper holding veto authority as the sole path to the process. **R2:** the gatekeeper's acceptance set is proven to be a subset of the truly-safe set over state, trajectory, and rate, such that no admissible sequence, ordering, or rate of admissible actions can compose into an unsafe path — a setpoint range check or a static bounds demonstration does not satisfy this. **R3:** any action that changes the pre-authorized envelope itself, or the gatekeeper's configuration, limits, or logic, or that moves the operating region toward a protective setpoint, is categorically excluded from advance discharge and takes in-the-moment authorization or offline requalification. Where the R2 proof cannot be produced, advance discharge is not available. The Command Broker discharges the authorization in advance by accepting the pre-authorized envelope and its R2 proof, and the gatekeeper then admits each action in the class; what the gatekeeper stands in for is the Broker's presence at the moment of action, not the Broker's judgment (*The Broker* §4.4). Advance discharge relocates the human authorization in time; it does not remove it, and it does not reclassify the action to below the line. An envelope against which no verified gatekeeper meeting R1 to R3 admits is not an authorization: it is at most a pre-evaluated basis for an in-the-moment authorization. The pre-approved action envelope is defined as an authorization type in CB-PAE (*Pre-Approved Action Envelope*), which assigns its governance to the facility's configuration management program and station operating procedures. The product requirements for the interface in which an envelope is composed are at §3.8, and for the supplier notification on which its validity depends at §3.9.

Two objects, not one. The consequence classifier required by §3.2 and the gatekeeper are two distinct objects in series and shall not be merged (FD-BL-D1 §5). The classifier answers placement and routes action by action; it is not a safety gate and does not authorize. The gatekeeper answers mode of discharge for actions already placed above the line; it vetoes and

does not classify. Classification occurs first. The evidentiary record shall show both the classifier's routing basis and the gatekeeper's acceptance-set proof.

DOWNGRADE PATHWAY (Should): The enforcement mechanism itself is not configurable — it shall not be defeasible by configuration change, software update, or system integration (FD-BL §4.5). What the product may offer is a documented transfer of the compliance obligation: a configuration option by which the AOO or integrator assumes responsibility for authorization of a stated class of actions. Activation must require a positive, deliberate configuration action — not a silent default, not a single-click override — and must generate a dated, auditable log entry identifying the action classes affected.

RISK TRANSFER OBLIGATION: When a downgrade is executed, documentation must state: (a) that Command Broker enforcement has been reduced from the DBA-series default; (b) the specific residual risk — actions the DBA series places above the Bright Line may reach physical command execution without the required authorization; (c) that the AOO assumes the compliance obligation; and (d) the AOO's identity and date of authorization.

Conformance and misrepresentation. Command Broker conformance is assessed by the deploying organization, not asserted by the vendor; a vendor's compliance claim is not a substitute for the AOO's assessment (CB-UC-1 §6). A correctly implemented interface that routes recommendations through a human who rubber-stamps every output is architecturally compliant and functionally worthless. Marketing the presence of a confirmation dialog as, in itself, Command Broker compliance — where the surrounding workflow, time allocation, or authority structure does not permit genuine independent evaluation — is a material misrepresentation of DBA-series compliance, distinct from and additional to any downgrade disclosed under this section.

Which instrument governs. This document governs the product specification. The Broker role itself — what kind of thing it is, the competence that qualifies a person to hold it, and the conditions under which a verified gatekeeper may stand in for its presence — is defined in FD-BR (*The Broker*), which governs. The CB-IB Qualification and Onboarding Standard carries the normative competency, onboarding, and accountability requirements, and the Command Broker Implementation Guide carries the asset owner's implementation obligations for establishing and sustaining the function. A vendor conforms to DBA-VC, not to those instruments, and is not asked to satisfy them independently. The vendor's obligation with respect to them is to supply the inputs the asset owner needs to discharge its own obligations — the vendor-supplied artifacts stated in §3.5 and, where the product discharges authorization in advance, the artifacts at §6.1 that the Command Broker requires when discharging in advance (FD-BL-D1 §7).

3.2 Bright Line Enforcement — Consequence Classification and Architectural Separation

DEFAULT (Must): The product architecture enforces the Bright Line mechanically, by classification rather than by a single gate. The product implements a consequence classifier that evaluates each action an in-scope component can produce and routes it on its placement: above-line actions to the authorization pathway required by §3.1, below-line actions to ordinary execution. An action is one change the component's output would make, if executed or acted upon, to a process parameter; to a protective function's settings, trip logic, bypass state or actuation criteria; to the pre-authorized envelope or the gatekeeper; or to an enterprise network the facility depends on, including isolating a segment, pushing a reimaging or shutting down a service, whose consequence may reach functions the network carries for others (FD-BL §4.3). The classifier evaluates each of these, and does not individuate an action more finely than the change it makes. The classifier's routing decisions, and the basis on which each action was classified, are part of the evidentiary record (FD-BL §4.3). Architectural separation of the advisory and command paths is where this enforcement is built; it is not the placement criterion, and a single gate at that seam does not discharge this requirement — it will leave above-line actions ungated on the advisory side and gate below-line actions unnecessarily on the command side. Both the classifier and the separation are verifiable through product documentation and independent testing. The enforcement mechanism shall not be defeasible by configuration change, software update, or system integration (FD-BL §4.5).

DEFAULT (Must), continued — classifier inputs. The state on which the consequence classifier routes is an input on which placement depends. Where that state is output of an in-scope component, it carries the indication-integrity obligation (FD-BL-D2 §4.1). The product shall make such inputs traceable to independently verifiable values (FD-BL-D2 §4.2). Where such an input cannot be corroborated through an independent path, the action it bears on shall not proceed without in-the-moment Command Broker authorization (FD-BL-D2 §4.3). That rule governs discharge and not placement: it does not place the action above the line, and the classifier shall record the action's placement as made on consequence, together with the uncorroborated input that withheld it from ordinary execution.

DOWNGRADE PATHWAY (Should): Where an integrator or AOO requires architectural modification that reduces the separation, the vendor must provide written guidance stating that such integration defeats the enforcement architecture the DBA series requires — which is not defeasible by integration (FD-BL §4.5) — and that the compliance obligation transfers to the integrator or AOO.

RISK TRANSFER OBLIGATION: Vendor product datasheets, integration guides, and API documentation must not represent an in-scope component as capable of "autonomous process optimization" or "direct control integration," implying autonomous authority over actions above the Bright Line, without a prominent disclosure that such capability is inconsistent with DBA-series compliance unless the Command Broker enforcement requirement in §3.1 is satisfied.

3.3 Component Scope and Placement Assessment — The Three-Question Test

Before deploying any AI component with a pathway to a physical process parameter, the asset owner must be able to answer three questions about the product. The vendor's documentation obligation is to make the answers available without requiring the AOO to conduct an independent investigation.

Question 1: Which actions can the component produce, and where is each placed? The vendor must document every pathway from the component to a physical process parameter, the consequence classification the product assigns to each action, and the basis for that classification. Placement is action-level; a single component will typically produce actions on both sides of the line (FD-BL §4.3).

Question 2: Is the component deterministic within a stated input bound and formally testable as deployed? The vendor must answer against that property test, not by product category, and must state the input bound where the carve-out is relied upon. Mixed products must assess each component independently (FD-LD §4.3).

Question 3: Has the in-scope component been subjected to red-teaming? Red-teaming must be conducted by parties independent of the vendor. The vendor must document scope, methodology, and findings.

DEFAULT (Must): Product documentation provides clear, unambiguous answers to all three questions and, where the product discharges any authorization in advance, to the FD-BL-D1 R1–R3 conditions at §3.1 — in each case without requiring the AOO to conduct an independent investigation.

DOWNGRADE PATHWAY (Should): Where documentation does not yet address all three questions, the vendor should provide a compliance roadmap with dated commitments for each element.

RISK TRANSFER OBLIGATION: Vendor representations about product safety or compliance that are inconsistent with the three-question answers create warranty and liability exposure.

3.4 Operator-Facing Advisory Workflow Constraints

The Command Broker works through two interface archetypes. The first is the live console, specified for Balancing Authority operations in CB-UC-1 (*BA Integrated Interface*), where the broker evaluates and authorizes a specific proposed action at operational tempo. The second is the deliberative composition interface at §3.8, where the responsible engineer and the broker compose an envelope in advance, at analytical tempo, for a bounded class of actions to be invoked later. The broker function is the same in both. The tempo differs, and so does the decision: in the first it is whether to take this action, and in the second it is where a class of actions can safely be bounded. The requirements of this section are console requirements. Several of the console's protections — a single operational view, a prohibition on navigating away, and a review-pause timer — would be wrong in composition and do not apply to it (*Pre-*

Approved Action Envelope §10). The same is true of information. The facility's configuration baseline is kept off the console because at operational tempo it is noise, and it is required in composition because there it is the substance of the work (*Pre-Approved Action Envelope* §3.2).

DEFAULT (Must): The product does not pre-populate control system inputs from the output of a component in scope under FD-LD §4.3. The product does not present such output as an actionable default for any action placed above the Bright Line. For those actions the workflow requires an independent verification step between the advisory output and the operator's authorization under §3.1, and the verification step is verification of the recommendation, not action upon it.

The product must also preserve the operator's independence from the advisory system, as CB-UC-1 R-IND-1 to R-IND-3 require. Those requirements are stated for Balancing Authority operations; they apply to every deployment with the sector's own process parameters in place of the Balancing Authority's, as CB-UC-1 §7 provides.

Read in plain language and in sector-neutral terms, R-IND-1 requires that the advisory display never be the operator's only source of process state, so that independent monitoring of the process stays continuously visible and is never obscured by the advisory interface. R-IND-2 requires that the operator be able to dismiss, minimize, or override any recommendation without consequence to operations, a dismissal not being an audit event and a confirmation being one. R-IND-3 requires that no recommendation be presented with more urgency than the operational situation warrants, its alarm treatment being configurable and defaulting below the urgency of the primary process alarms. These readings are an aid to the reader; CB-UC-1 states the requirements and governs.

DEFAULT (Must), continued — indication integrity. Where advisory output can function as the basis for an above-line authorization — whether by a human Command Broker or by a verified gatekeeper — it carries the indication-integrity obligation, whatever its own placement (FD-BL-D2 I1–I3). The product shall make that output traceable to independently verifiable values: values confirmable through a path independent of the component that produced the output — a direct read of the underlying measurement, a diverse or redundant instrument, or a deterministic sensor channel — such that a fault, drift, or manipulation in the synthesis does not also corrupt the confirming path (I2). Output that the producing component both generates and certifies is not independently verifiable. The product shall not permit such output to be the sole basis for an above-line authorization: an independent confirming path shall be available to the authorizer before the authorization is valid, and where no such path exists the authorization is unavailable (I3). The product shall specify its denied-indication behavior: where the confirming path is unavailable, or the synthesis and the independent channel diverge beyond a stated tolerance, above-line authorization on that basis shall be disabled, and re-establishment shall require a deliberate, authenticated act. Correctness is necessary and not sufficient. Where the basis for an above-line authorization is correct but does not show what the authorization depends on, the product shall surface what is missing before the authorization is relied on, and an

authorization given on a basis the product has flagged as incomplete shall be recorded as given on an incomplete basis (FD-BL-D2 §4.3). How what is missing is identified is open in FD-BL-D2 and is not settled here.

DOWNGRADE PATHWAY (Should): Where AOO workflow design reduces the independence of the verification step, the vendor's integration guidance must flag this as an indication-integrity risk under FD-BL-D2 I3 and recommend corrective workflow design. The corroboration requirement at I3 is a hard condition; a sector annex may set its grade but may not waive it.

RISK TRANSFER OBLIGATION: Cloud-connected advisory tools that have any pathway reaching the OT network create a network boundary risk independent of the Command Broker concern. Vendor documentation must disclose any network connectivity pathways that could reach OT systems.

3.5 Model Provenance and Supply Chain Assurance

DEFAULT (Must): The vendor maintains and provides to AOOs upon request: documented verification of training data sources; version control with change logging for model updates; vendor certification of update integrity; and a defined rollback process allowing the AOO to revert to a prior model version.

DEFAULT (Must), continued: The vendor also provides (e) an operator-facing AI system operational profile summary — process domain, training scope, and known limitations — in a form the Command Broker can read and apply during evaluation and qualification; (f) proactive advance notification to the AOO of any model update, fine-tuning change, or change to retrieval-augmentation sources, in a form and on a timeline that lets the AOO complete a broker requalification review before the changed system is placed in front of a broker; and (g) a qualification exercise mode that lets the AOO inject known-erroneous recommendations for broker qualification and continuing-competency exercises without affecting the production process. Items (e) to (g) are the vendor-supplied inputs on which CB-CT_Commissioning_and_Joint_Training and the Command Broker Implementation Guide rely for broker qualification, joint training and requalification.

DEFAULT (Must), continued — vendor releases: A model, firmware or configuration release the vendor makes to its installed base is not by itself an action under FD-BL §4.3; the controls this section places on the vendor are obligations the AOO places on the vendor by contract, derived from the facility's design basis. At the facility, taking in a release that would change a control or protective function is an action placed on its consequence, and the product shall admit it only through a governed act (Local Sovereignty §3.2), never by automatic update. Where a release is taken in by stages, each widening is its own action, and the first stage is placed on the consequence of the whole planned reach. A release that changes a carved-out component's tested

configuration ends the carve-out for that component until the new configuration is formally tested (FD-LD §4.3).

DEFAULT (Must), continued — the system nameplate. A generator, a turbine, or a blade server carries a nameplate tied to procurement engineering, from which the detailed specification can be found. A Gen AI system has no equivalent, and the gap matters more for it than for any of them, because what the facility deploys is a composite whose behavior can change when any one of its parts changes. The vendor therefore issues a nameplate for each deployed configuration, and maintains it. The nameplate identifies the deployed system, not the model alone. It carries a single nameplate identifier, together with the elements that identifier stands for: the foundation model, by name, version and supplier; any fine-tuning or adapter, by identifier and version; a reference to the provenance record of the training data; the version of the system prompt; the set of retrieval or context sources and its version; the set of tools the component can call and its version; and the deployment target, whether on premises or supplier-hosted, with its location. The nameplate identifier changes whenever any element changes. The provenance record behind the nameplate is the supplier's intellectual property and is made available to the asset owner under the confidentiality terms of the procurement instrument, as the hybrid quality-assurance method already provides for (*Formal Methods for Trustworthy GenAI*, Appendix C). The vendor is not required to disclose model weights. It may place a cryptographic digest of the weights on the nameplate, which shows that the weights have not changed without revealing anything about them; this is optional, and where it is supplied it is the only element by which the facility can detect a supplier-side model change without relying on the supplier's notice. The nameplate may be issued in either of the two established machine-readable bill-of-materials formats, SPDX 3.0 (with its AI and Dataset profiles) or CycloneDX (with its machine-learning model component and model card). The identity, dataset and digest elements use those formats' own fields; the system prompt, retrieval sources, tool set and deployment target, for which neither format has a field, are carried as extension properties under the names this section gives them. The nameplate identifies a configuration and not a behavior. Two systems carrying the same nameplate can behave differently in service, and that difference is the province of surveillance, not identification (*Pre-Approved Action Envelope* §4).

DOWNGRADE PATHWAY (Should): Where the vendor's model update process does not yet include all four provenance elements, the vendor should provide a phased implementation roadmap, prioritizing integrity certification and rollback capability.

RISK TRANSFER OBLIGATION: Model updates delivered without vendor certification of update integrity place the AOO in the position of deploying an unverified change to a highest-criticality application. The vendor's update release documentation must state whether integrity certification is provided.

3.6 Red-Teaming Obligations

DEFAULT (Must): The vendor has subjected each component in scope under FD-LD §4.3 that can produce an action placed above the Bright Line to red-team assessment by a party independent of the vendor, covering at minimum: prompt injection attacks; training data poisoning; adversarial inputs designed to produce unsafe process parameter recommendations; and bypass of the Command Broker confirmation step.

DOWNGRADE PATHWAY (Should): Where independent red-teaming has not yet been completed, the vendor should clearly mark the affected components as pending red-team validation and provide a dated commitment to complete the assessment.

RISK TRANSFER OBLIGATION: Vendor representations that an in-scope component is suitable for deployment where it can produce above-line actions without a documented independent red-team assessment create a material misrepresentation risk.

3.7 Audit Record Requirements

Every Command Broker confirmation, and every below-Bright-Line confirmation, produces an audit record. The record is the instrument by which a reviewer, a regulator, or an incident investigator later reconstructs why a broker acted, not merely that they did. This requirement applies CB-UC-1 §5 in sector-neutral form, with one strengthening. CB-UC-1 R-AUD-1 admits a documented schema as one of several satisfying open formats; this document makes the documented schema mandatory, because an open format without a published schema is not independently parseable by the compliance staff for whom R-AUD-2 reserves access.

DEFAULT (Must): The product satisfies CB-UC-1 R-AUD-1 to R-AUD-4 and R-AL-4 for every confirmation, with the documented schema mandatory as stated above. Where an authorization is discharged in advance under §3.1, the record shall additionally resolve to the named Command Broker who authored and accepted the pre-authorized envelope, discharging in advance, and shall capture the gatekeeper's disposition of each proposal (FD-BL-D1 §8).

DEFAULT (Must), continued — the envelope chain. Where the product supports envelope authorization, each envelope's composition record (§3.8), approval record and invocation record are bound to that envelope and to one another. The chain reads composed, approved, invoked, and each link is recoverable from the next (*Pre-Approved Action Envelope* §7.4). The approval record is the facility's configuration change package and procedure change, held by the asset owner; the product carries its identifier. The write-before-proceed rule of CB-UC-1 R-AL-4 applies at both ends: an envelope is not available for invocation until its approval record is written, and an invocation does not execute until its record is written. The invocation record identifies the envelope, its version, and its status at the moment of invocation.

DOWNGRADE PATHWAY (Should): Where the product's audit store is not yet independent of the advisory component, or not yet in an open documented format, the vendor provides a phased roadmap prioritizing independent, AOO-accessible storage and an open export format.

RISK TRANSFER OBLIGATION: When audit records are retrievable only through the vendor's own platform or tooling, documentation must state: (a) that the independent-access requirement (R-AUD-2) is not satisfied; (b) that the AOO cannot independently reconstruct broker decisions without vendor cooperation; and (c) the AOO's acknowledgment and date of authorization. A vendor whose audit trail lives solely inside its own platform does not satisfy this requirement.

3.8 Deliberative Composition Interface

An envelope is composed by the responsible engineer and the Command Broker together, and the authorizing act is the broker's, discharging in advance (*The Broker* §4.2; FD-BL-D1). The composition interface is where that work happens, and its design decides whether the record it leaves shows that an independent evaluation took place. Its lineage is the analyst's workstation of the Analyst's WorkBench, where the analytical streams behind the glass are integrated on one screen so the analyst sees the whole of an issue. In that design the reviewer moves material between panes by drag and drop; that is offered here as an illustration of the archetype, not as a requirement.

DEFAULT (Must): The product provides an integrated composition workspace in which the engineer and the broker assemble the picture themselves from multiple analytical streams, rather than selecting from options the system presents. The workspace presents the facility's configuration baseline — the current configuration, the out-of-service list, the temporary modifications in force, and the changes made since the in-scope component's picture of the plant was established — as an input to composition (*Pre-Approved Action Envelope* §3.2). The configuration state and the system nameplate identifier (§3.5) at the time of composition are recorded as part of the envelope's basis, so the envelope can later be re-validated against the state it was composed in.

DEFAULT (Must), continued — interaction log and reasons. The product generates an interaction log of each composition session, and the log is the composition record (*Pre-Approved Action Envelope* §7.1). It records what the engineer and the broker opened, from which analytical source, in what order, and which alternatives they weighed. The product writes it as the session proceeds, not reconstructed afterward, to the audit store under §3.7. For each bound the envelope sets, the broker records why the bound falls where it does, by selecting from a controlled list of reasons. The list belongs to the asset owner, is maintained under its configuration management program, and changes only through that program. The vendor supplies the mechanism and may supply a starting list, but does not own the content. Where no listed reason fits, or where the bound departs from what the in-scope component proposed, the broker adds a brief free-text statement, and the product does not accept the envelope without it. A log shows what the broker looked at but cannot show why; the controlled list keeps that record short enough for every broker to write and uniform enough to review. The reasoning belongs to the asset owner's configuration management record, not to the product: the interaction log and

the reason record enter the basis carried by the approval record and are retained with it (*Pre-Approved Action Envelope* §7.1, §7.2).

DEFAULT (Must), continued — indication integrity. Composition does not relax indication integrity. Where an envelope's bound rests on output of an in-scope component, that output carries the obligation set by FD-BL-D2 (*Indication Integrity*). It must be traceable to values that can be confirmed through a path independent of the component that produced it, so that a fault or manipulation in the component cannot also corrupt the check on it, and it may not be the sole basis for the bound. The confirming path at composition may be slower and more thorough than the console's, but it must exist (§3.4).

DEFAULT (Must), continued — envelope status. The product carries each envelope's status on the envelope itself — available, suspended, pending re-validation, restored — and the invocation check reads that status before admitting any action under the envelope. Suspension takes effect when the triggering change is entered. Restoration is an affirmative act by the same review authority that approved the envelope, exercised through the facility's configuration management program, which governs these actions; the product provides no other path to restore an envelope (*Pre-Approved Action Envelope* §5.4). Where the dependency between a change and an envelope is uncertain, the product treats the envelope as affected.

DOWNGRADE PATHWAY (Should): Where an asset owner captures the composition record outside the product, for instance in its own configuration management system, the vendor's documentation describes how the external record is bound to the envelope. Activating that arrangement is a positive, documented configuration act that generates a dated log entry. The obligation to produce a complete composition record then rests with the asset owner. The interaction log, the reason record and the envelope status are relocated by the transfer, not waived.

RISK TRANSFER OBLIGATION: Where the product can produce an envelope without an interaction log, documentation must state: (a) that such an envelope carries no record that an independent evaluation took place; (b) that it therefore cannot support an advance discharge under §3.1 and is at most a pre-evaluated basis for an in-the-moment authorization (*Pre-Approved Action Envelope* §7.1); and (c) the asset owner's acknowledgment and date of authorization.

3.9 Supplier Model-Change Notification

An envelope is composed and approved against the observed behavior of a particular system. That behavior can shift through a model update, a retraining cycle, a change to retrieval or context, or a supplier-side deployment, none of which reaches the facility looking like a design change. If the envelope is a controlled item, a supplier model revision is an unreviewed change to a controlled item (*Pre-Approved Action Envelope* §5.1). CB-PAE establishes the obligation

and assigns its product-specification form to this document (*Pre-Approved Action Envelope* §5.2). This section supplies that form, using the system nameplate at §3.5 as the unit of identity.

DEFAULT (Must): A material change is any change to the deployed system that changes its nameplate identifier under §3.5. The procurement instrument states that definition and may add further classes of change to it. It may not narrow it. The vendor notifies the asset owner of every material change in advance of deployment, giving the prior and the new nameplate identifiers and the elements that changed. The product records, for each envelope, the nameplate identifier it was composed against. A new configuration does not displace the one an envelope was composed against until the envelope has been carried across to it. Until then, invocations under the envelope run against the prior configuration, and the vendor keeps that configuration available for the purpose until cutover. An envelope is carried across in one of two ways: by individual re-validation under §3.8, or by an equivalency acceptance under the paragraph that follows. On receipt of a notification, the product identifies every envelope composed against a prior identifier and holds it bound to that identifier until one of the two is complete (*Pre-Approved Action Envelope* §5.2). Where the nameplate carries a weights digest, the product checks the digest of the running system against the nameplate, and treats a mismatch as a material change that has not been notified.

DEFAULT (Must), continued — equivalency acceptance. The facility's configuration management program may show, by a single acceptance evaluation, that a new nameplate is equivalent to the one it updates for the envelopes the evaluation names. It may then carry those envelopes across together rather than one by one. This is the practice by which a nuclear plant accepts a replacement item that is not identical to the original. One engineering evaluation shows that the replacement meets the functional requirements the original met, and the applications that depend on it keep their basis. For a Gen AI system the equivalence is one of behavior, not of parts. A new nameplate differs from the old by definition, so the evaluation must show that the new configuration behaves the same with respect to what each named envelope depends on (*Pre-Approved Action Envelope* §5.2).

The evidence has two halves. The vendor supplies a description of the change for each nameplate element that moved, and the vendor's regression results. The asset owner runs a defined acceptance test set against the new configuration, drawn from the composition records of the envelopes in question and from the cognitive error taxonomy on which the tiered assessment method rests. A vendor statement that nothing material changed is not an evaluation. Because the system is generative, the evaluation is empirical: it shows that the new configuration meets a stated acceptance criterion on the behavior tested, and does not prove equivalence in the sense a formal verification would. The evaluation therefore states its acceptance criterion before the test set is run, and the product records the criterion, and when it was fixed, ahead of the test results.

Coverage is stated envelope by envelope. For each envelope it covers, the evaluation names the nameplate elements on which that envelope's basis depends and the evidence that the behavior it depends on is unchanged. An envelope whose basis rests on an element that changed is excluded

and is re-validated individually. Examples are a retrieval source the envelope relied on, or a tool through which its actions pass; a new tool the component can call opens pathways no envelope was bounded against. An evaluation that lists the envelopes it covers without stating why each keeps its basis is a bulk restoration, and the product does not re-bind envelopes on it (*Pre-Approved Action Envelope* §5.2, §5.4).

The evaluation is approved by the same review authority that approves envelopes, through the configuration management program. On approval, the product re-binds each covered envelope to the new nameplate identifier, recording both identifiers and the evaluation that links them as part of the envelope's basis, so the chain at §3.7 is unbroken. Re-binding does not reset an envelope's expiry, which continues to run from the envelope's own approval (*Pre-Approved Action Envelope* §4.4). The product records each accepted equivalency evaluation as a trigger for the aggregate review of the envelope set (*Pre-Approved Action Envelope* §6). Cutover to the new configuration is a governed act of that program. The prior configuration is retired only after every envelope bound to it has been carried across or withdrawn.

DOWNGRADE PATHWAY (Should): Where a vendor's service model includes supplier-side changes that cannot be notified in advance, the vendor identifies those change classes in the procurement instrument, and the asset owner's acceptance of them is a documented act. The vendor still issues a new nameplate identifier for each such change and notifies it on deployment, so that the envelopes it reaches can be suspended. Where a vendor cannot keep a prior configuration available until cutover, the envelopes bound to it are suspended on deployment of the new configuration, and invocation reverts to per-action Command Broker authorization until they are carried across.

RISK TRANSFER OBLIGATION: When a supplier keeps the right to change the deployed system without issuing a new nameplate identifier, documentation must state: (a) that envelopes composed against the system cannot be shown to remain valid after such a change; (b) that the asset owner should treat every envelope as pending re-validation after any unnotified change it detects; and (c) the asset owner's acknowledgment and date of authorization.

Traceability. Each canonical Part I requirement is governed by a section of the Command Broker corpus, as follows:

Canonical requirement (Part I)	Governing CB instrument section
§3.1 Command Broker Enforcement	FD-BL §4.5 and *The Broker* §4.1 (governing); FD-BL-D1 R1–R3 (advance discharge); CB-Framework §3.1–3.2; CB-IB Qualification Standard §2 (accountability principle)
§3.2 Bright Line Enforcement	FD-BL §4.2–§4.3 (governing); FD-BL-D2 §4.1–§4.3 (classifier inputs); CB-Framework §3.1, §3.2a (consequence classifier)
§3.3 Scope and Placement Assessment	CB-IB Qualification Standard §4.1 (competency domains)
§3.4 Operator-Facing Workflow Constraints	FD-BL-D2 I1–I3 and §7 (governing); CB-UC-1 §4.5 (R-

	IND); §4.2–4.3
§3.5 Model Provenance & Supply Chain	Command Broker Implementation Guide §5.2, §8 (provenance documentation; update-triggered requalification)
§3.6 Red-Teaming	Command Broker Implementation Guide §7 (adversarial-realism exercises); CB-IB Qualification Standard §4.1 (adversarial scenario awareness)
§3.7 Audit Record Requirements	CB-UC-1 §5 (R-AUD-1…4) and §4.3 (R-AL-4); CB-IB Qualification Standard §7; *Pre-Approved Action Envelope* §7.4 (envelope chain)
§3.8 Deliberative Composition Interface	*Pre-Approved Action Envelope* §3.2, §5.4, §7.1, §7.2, §10
§3.9 Supplier Model-Change Notification	*Pre-Approved Action Envelope* §4.4, §5.1–§5.2, §6

4. Sector Applicability Matrix

The following table maps each canonical requirement in §3 to the sector frameworks where it appears. “Anticipated” indicates that the requirement is established as a series standard and will be incorporated into the named sector framework on its next substantive revision.

Requirement	DBA-W Water	DBA-GAS	DBA-OIL	DBA-ES Electric	DBA-NUC-SF Nuclear	EMS (CntlCo)
3.1 Command Broker Enforcement	§5.5.2	§5.4	§5.4	§3.4 (anticipated)	Safety system I&C interposition; SRO as locus of the Command Broker function, qualified per CB-Annex-Nuclear	§7.1.2 (licensed operator mandate)
3.2 Bright Line Enforcement	§5.5	§5.4	§5.4	§3.4 (anticipated)	Safety/non-safety isolation as enforcement architecture; placement per CB-Annex-Nuclear	§7.1.1; §7.2

3.3 Three-Question Assessment	§5.5.3	§5.4	§5.4	Anticipated	Fourth question: software qualification	§7.1.3 (fourth question: adversarial injection)
3.4 Workflow Constraints	§5.5.4	Anticipated	Anticipated	Anticipated	Anticipated	§7.1.2 (role-based enforcement)
3.5 Model Provenance	§5.5.5	Anticipated	Anticipated	Anticipated	Software qualification pedigree	Anticipated
3.6 Red-Teaming	§5.5.3	§5.4	§5.4	Anticipated	IV&V requirement analog	§7.1.3 (adversarial injection testing)
3.7 Audit Record Requirements	§5.5 (anticipated)	Anticipated	Anticipated	Anticipated	10 CFR 50 App. B records	§7.1.2 (CB confirmation record)
3.8 Composition Interface	Anticipated	Anticipated	Anticipated	Anticipated	10 CFR 50.59 change evaluation as approval authority (*Pre-Approved Action Envelope* §3.5)	Anticipated
3.9 Model-Change Notification	Anticipated	Anticipated	Anticipated	Anticipated	Commercial-grade dedication analog (*Pre-Approved Action Envelope* §5.3)	Anticipated

5. The Procurement Signal

The AI Governance Series Overview §6.1 states the market reality directly: the pace of AI deployment in critical infrastructure is outrunning the standards process, but the standards process is moving. NERC is aware of the gap. NRC is aware of the gap. ISA and IEEE are actively developing standards. Vendors who build to the Bright Line before it becomes mandatory are building a qualification record and a product architecture that will survive the regulatory transition.

5.1 The Propagation Cascade

Step 1 — Series publication and engineering firm adoption. Engineering and consulting firms that serve critical infrastructure sectors adopt the DBA framework as their internal assessment standard. When a firm uses DBA-series event categories in a client assessment, it is applying the DBA standard without a regulatory mandate.

Step 2 — Utility procurement language. Asset owners whose assessments have been conducted against DBA-series standards begin incorporating DBA language into their procurement specifications. This happens across sectors near-simultaneously as the series matures.

Step 3 — Vendor product differentiation. Vendors who have built to the DBA-series specification can respond to procurement language immediately. Vendors who have not must retrofit — under procurement deadline pressure, across multiple sectors simultaneously.

Step 4 — Regulatory incorporation. The DBA series is designed for regulatory adoption. The water sector pathway runs through EPA's AWIA resilience assessment guidance. The electric sector pathway runs through NERC reliability standards. The nuclear sector pathway runs through NRC rulemaking.

5.2 The Default/Downgrade Structure as a Market Signal

A vendor whose product ships with Command Broker enforcement on by default is telling the market: we built to the standard. A vendor whose product ships with enforcement off by default, or who offers enforcement as an add-on module, is telling the market: we built to a different standard. As procurement language matures, the default state question will become a standard due diligence item in vendor evaluation.

5.3 The Vendor Compliance Narrative

Vendors who build to the DBA-VC specification have a specific compliance narrative available:

- Our in-scope components ship with Command Broker enforcement active by default. No configuration action by the AOO is required to achieve DBA-series alignment.
- We provide documented answers to the three-question assessment (§3.3) in our standard product documentation.
- We have conducted an independent red-team assessment of our in-scope components against sector-specific attack scenarios.
- We maintain model provenance records and provide rollback capability for all in-scope components in OT applications.
- When an AOO or integrator requests a downgrade, we provide documented risk transfer records that make the AOO's compliance obligation explicit.

This narrative is not marketing language — it is an evidentiary claim. Each element is verifiable by an AOO, an engineering firm, a regulator, or an incident investigator.

6. Path to Compliance Demonstration

No DBA-series certification program currently exists. What vendors can do is assemble an evidentiary package that supports a credible compliance claim now and positions the organization to transition to formal certification when a program is established.

6.1 The Evidentiary Package

Product Architecture Documentation. A document describing the Bright Line enforcement mechanism — how it is implemented, how it is verified, and what the default configuration state is. Sufficient detail to allow independent assessment without vendor-guided demonstration.

Component Scope and Action Placement Inventory. A document listing each AI component, the determinism and stated-input-bound finding that puts it inside or outside scope under FD-LD §4.3, and, for each in-scope component, the set of actions it can produce with the placement assigned to each and the basis for that placement. Updated at each product release.

Three-Question Assessment Results. Vendor-generated answers to the three questions in §3.3, with supporting evidence.

Independent Red-Team Report. A report from a party independent of the vendor. Available to AOOs under NDA where commercially sensitive findings are included.

Model Provenance Records. Documentation of training data sources, version history, update integrity certification methodology, and rollback procedure, and the system nameplate for each deployed configuration, in the format supplied (§3.5).

Downgrade Configuration Procedure. A procedure document describing the steps required to execute a downgrade, including the configuration log entry generated and the risk transfer documentation produced.

AI System Operational Profile Summary. An operator-facing document stating the process domain, training scope, and known limitations of each in-scope component, suitable for the Command Broker to apply during evaluation and qualification.

Model Update Notification Procedure. The procedure and timeline by which the vendor notifies the AOO of model updates, fine-tuning changes, and retrieval-source changes in advance of deployment, and the governed act by which the AOO takes each release in, stage by stage.

Qualification Exercise Mode Description. Documentation of the product's mode for injecting known-erroneous recommendations to support broker qualification and continuing-competency exercises.

Commissioning sequence. These vendor-supplied artifacts — the operational profile summary, the model-update notification procedure, and the exercise mode — are placed in an ordered commissioning and re-qualification sequence, with joint training and gate sign-off, by the CB-CT Commissioning & Joint Training module (CB-CT_Commissioning_and_Joint_Training).

Gatekeeper Acceptance-Set Proof. For any action class the product discharges in advance under §3.1, the R2 completeness and soundness proof over state, trajectory, and rate; the identity and qualification basis of the Command Broker who authored and accepted the pre-authorized envelope, discharging in advance; and the R3 change-control mechanism by which changes to the envelope, or to the gatekeeper's configuration, limits, or logic, are excluded from the advance pathway. Where the product discharges no authorization in advance, a statement to that effect.

Indication-Integrity Design Statement. For each output that can function as the basis for an above-line authorization: the independently verifiable values to which it is traceable, the confirming path and its independence of failure from the producing component, the divergence tolerance, and the denied-indication behavior (§3.4; FD-BL-D2 I2–I3).

Audit Record Specification. The audit record schema, storage-independence design, tamper-evidence mechanism, retention period, and the AOO-independent access method (per §3.7).

Composition Interaction Log Specification. The schema of the interaction log and of the reason record; the controlled-list mechanism and any starting list supplied; how the composition, approval and invocation records are bound to one envelope; and the envelope status model with its suspension and restoration paths (§3.7, §3.8).

Envelope Version Binding Record. The materiality definition in force under the procurement instrument, the notification procedure against it, and the method by which the product identifies, for each envelope, the nameplate identifier and configuration state it was composed against (§3.9).

Equivalency Acceptance Support. The per-element change description and regression evidence the vendor supplies with each material change; the mechanism by which the product holds envelopes bound to a prior nameplate identifier, re-binds them on an accepted equivalency evaluation and records both identifiers; the vendor's arrangements for keeping a prior configuration available until cutover; and the mechanism by which the product records the acceptance criterion ahead of the test results, carries each envelope's expiry unchanged across re-binding, and flags the aggregate review (§3.9).

6.2 Engagement Pathway

Vendors who wish to engage with the DBA series development process — to provide input on sector-specific product requirements, to participate in the development of FC instrument documents, or to be positioned as reference implementations for the Command Broker architecture — should contact Eclectic Technologies directly. The series is in active

development. Vendor input on the translation from Design Basis requirement to product specification is valuable and welcome.

PART II — SECTOR CHAPTERS

7. Introduction to Sector Chapters

Part II instantiates the cross-sector foundation of Part I for each sector in which DBA instruments are active. Each chapter identifies the sector-specific facility classes, the highest-criticality application contexts, and the product requirements that go beyond the canonical requirements in §3 due to sector-specific operational or regulatory conditions. Vendors serving multiple sectors should read all applicable chapters and treat the most stringent requirement as the floor.

Part II currently contains substantive chapters for the two most analytically mature DBA sectors: Energy (§8) and Nuclear (§9). Framework stubs are provided for Water, Oil and Gas, and other sectors where DBA work is in progress. Those stubs will be expanded in subsequent revisions as the applicable DBA sector frameworks mature.

8. Energy Sector (DBA-EN)

8.1 Sector Overview

The DBA-EN Energy Sector series is the most extensively developed DBA sector series. It encompasses three sub-sectors — electric (DBA-ES), natural gas (DBA-GAS), and petroleum (DBA-OIL) — each with its own sector framework and facility-class instruments. The governing parent document is DBA-EN-SF, which establishes the DB→DBA→DBA-MA three-layer progression as mandatory across all sub-sector series.

The energy sector presents the cross-sector amplifier argument with particular force. A vendor supplying an ICS platform for a pipeline compressor station may be selling to a facility that is simultaneously subject to DBA-GAS (natural gas sub-sector) and DBA-ES-GC-FC4 (the gas-electric cascade facility class, which propagates consequences through the electric sector). The Command Broker and Bright Line requirements apply at both the gas and electric consequence chain levels.

8.2 Highest-Criticality Applications — Energy Sector

Facility Class	Highest-Criticality Applications	Governing DBA Instrument
Balancing Authority / RC /	State estimator ML augmentation with AGC	DBA-ES-CntlCo §7

TOP (CntlCo)	pathway; Gen AI operator decision support; bad data detection, adversarial robustness	
EHV Transmission Substation (FC2)	Protection relay logic; automatic reclosing; transformer protection systems	DBA-ES-TX-FC2
HVDC Interconnect (FC3)	Converter control systems; power flow management; automatic voltage regulation	DBA-ES-TX-FC3
Gas Compressor Station (GC-FC4)	Compressor control and shutdown logic; pipeline pressure management; gas-electric cascade coordination	DBA-ES-GC-FC4
Renewable Generation / BESS (GN-FC8)	Inverter control firmware; BESS charge/discharge management; grid-forming functions	DBA-ES-GN-FC8
Natural Gas Transmission (GAS-FC1)	Compressor station operations; pipeline pressure management; SIS parameters; ESD logic	DBA-GAS-FC1
Natural Gas Processing (GAS-FC2)	APC/RTO optimization-safety boundary; SIS trip point setbacks; NGL separation process parameters	DBA-GAS-FC2
Petroleum Refining (OIL-FC1)	DCS process parameters; HF alkylation safety controls; SIS functions	DBA-OIL-FC1

8.3 Energy Management System — CntlCo-Specific Requirements

This section translates the AI governance interaction points established in DBA-ES-CntlCo §7 into product specifications for energy management system (EMS) vendors. These three interaction points are series firsts: they do not appear in any other DBA-ES instrument and carry design obligations specific to EMS products serving Balancing Authority (BA), Reliability Coordinator (RC), and Transmission Operator (TOP) functions.

8.3.1 State Estimator ML Augmentation — QA Obligations and the Gen AI/ML Layered Architecture

The Bright Line is a consequence threshold, not a technology classification. Actions whose erroneous execution could produce a BES reliability event are above the Bright Line; actions whose erroneous execution produces a correctable, facility-local consequence are below it. For state estimator ML augmentation in the EMS context, this distinction governs where each action a component produces is placed, and which mode discharges the authorization for those placed above the line.

An ML-augmented state estimator — bad data detection, topology processing, observability analysis — that is deterministic within a stated input bound and formally testable as deployed is

outside the Command Broker obligation under the carve-out (FD-LD §4.3). That is a scope finding, not a grant of ungoverned autonomy, and it does not place the estimator's actions below the Bright Line: their actions are among the highest-consequence available in the CntlCo environment, and the carve-out is defensible only because their behavior is bounded by proof. The governing obligation is accordingly not advisory-only output but QA rigor sufficient to sustain that bound: extensive testing, red-teaming against adversarial injection scenarios, and documented performance specifications are required before closed-loop operation is permitted. The ML system carries the determinism guarantee; the QA process validates that the guarantee holds at the operating limit.

Gen AI may participate in the state estimation and dispatch advisory chain, but only through a layered architecture: the Gen AI component holds proposal authority only and passes a command to the ML implementation layer, which admits the command only where it lies inside the layer's proven acceptance set and implements it if so. The Gen AI component never touches a physical process parameter directly. This architecture does not change where the Bright Line falls for any action — placement remains a function of consequence, action by action, and classifying the composite rather than the generating component is foreclosed (FD-BL-D1 §6). What the architecture can supply, for a narrow class of above-line actions, is admission in place of the Command Broker's presence at the moment of action, and not the authorization itself. The Command Broker discharges the authorization in advance, by accepting the pre-authorized envelope and its proof, and the ML implementation layer then admits each action in the class, only where it qualifies as a verified gatekeeper under FD-BL-D1 R1–R3 (*The Broker* §4.4): sole path to the process, non-bypassable and non-modifiable by the proposer (R1); acceptance set proven complete and sound over state, trajectory, and rate rather than checked against a parameter range (R2); and any command that would alter the layer's own validated parameters, limits, or logic categorically excluded from this pathway and routed to in-the-moment Command Broker authorization or offline requalification (R3). Where the R2 proof has not been produced, the layered architecture is not a discharge and the action takes in-the-moment authorization. Where a Gen AI-initiated command would produce an above-Bright-Line consequence for which no valid pre-authorized envelope exists — automatic redispatch, AGC setpoint modification, load shedding — the Command Broker intercepts before the command reaches the ML implementation layer.

The QA specification for ML components operating closed-loop under the carve-out must separately address two distinct performance dimensions. The first is random error tolerance: bad data detection algorithm performance against random measurement errors, the standard specification that vendors currently provide. The second is adversarial injection tolerance: algorithm performance against coordinated false data injection at multiple measurement points — the scenario in which an adversary crafts inputs to cause the ML system to produce a confident but false system state picture without triggering bad data detection. These are structurally different attack surfaces. A specification that addresses only random error tolerance does not satisfy the QA obligation for closed-loop autonomous ML operation in the CntlCo

environment. This adversarial injection requirement is grounded in DBA-ES-CntlCo §7.1.3, which establishes that the MA-1 governing DBA-MA scenario — state estimator corruption via false data injection — is structurally identical to an adversarial ML attack on the state estimator's inference inputs.

DEFAULT (Must): The EMS product ships with a documented QA package for each ML component operating closed-loop under the carve-out. The package must separately specify: (a) random error tolerance — bad data detection algorithm performance against random measurement errors; and (b) adversarial injection tolerance — performance against coordinated false data injection at multiple measurement points. A single combined tolerance figure does not satisfy this requirement. Closed-loop ML operation under the carve-out is not permitted without both specifications on record. Gen AI components that pass commands to the ML implementation layer must have an architecturally enforced consequence classifier: commands placed below the Bright Line may be passed to the ML layer for execution within its validated parameters; commands that would produce above-Bright-Line consequences are routed to in-the-moment Command Broker authorization before reaching the ML layer, unless the ML layer qualifies as a verified gatekeeper under FD-BL-D1 R1–R3 and the command falls inside its proven acceptance set. The QA package shall additionally document the independently verifiable path by which the estimator's output can be corroborated (FD-BL-D2 I2–I3) — the diverse or redundant measurement channel, its independence of failure from the estimator, and the cross-check cadence — together with the divergence tolerance beyond which above-line authorization on that state picture is disabled. Adversarial injection tolerance testing measures how far the estimator can be pushed; it does not supply the anchor, and a tolerance figure is not a substitute for an independent confirming path.

DOWNGRADE PATHWAY (Should): Where an AOO deploys ML components without a complete QA package — including legacy state estimator deployments — the component does not meet the carve-out and is in scope under FD-LD §4.3 until the package is complete. Its actions are placed on consequence, and those above the line route to the Command Broker under §3.1. The vendor should provide a compensating specification and a dated commitment to complete the full package; the compensating record governs what is on file during the interim, not whether the obligation applies. A compensating package does not substitute for a full QA package; it is a documented interim authorization with a dated commitment to complete the full specification. Where the vendor's adversarial injection tolerance assessment has not yet been completed, the vendor should provide a phased roadmap with a dated commitment.

RISK TRANSFER OBLIGATION: When a Gen AI component has a pathway to the ML implementation layer without a consequence classifier enforcing the above/below-Bright-Line routing, documentation must state: (a) that the consequence-classification gate required under this section is not in place; (b) that Gen AI-initiated commands may reach ML execution for above-Bright-Line consequences without Command Broker interposition; and (c) the AOO's acknowledgment, the compensating controls in place, and the date of authorization. Vendor

representations that an EMS's bad data detection capability satisfies DBA-series requirements for CntlCo applications must be supported by both the random error tolerance and adversarial injection tolerance specifications. A representation based only on standard bad data detection performance does not constitute DBA-ES-CntlCo §7.1.3 compliance.

8.3.2 Gen AI EMS Operator Decision Support — Action-Specific Bright Line Placement and Licensed Operator as Command Broker

The Bright Line placement for a Gen AI operator decision support tool is action-specific, not tool-specific. The same EMS advisory system may present below-Bright-Line options — incremental economic dispatch within an existing operating range, battery dispatch within a defined operating band — and above-Bright-Line options — involuntary load shedding, actions affecting multiple BAs, irreversible system state changes — in a single operational view. The Command Broker requirement governs the above-Bright-Line options. It does not govern the tool's presence in the environment or its use for below-Bright-Line advisory functions.

For the CntlCo environment, DBA-ES-CntlCo establishes a specific Command Broker constraint: the Command Broker for any above-Bright-Line action must be the licensed system operator on duty. This follows from the NERC system operator certification standard and the operator's legal accountability under NERC's rules of procedure. The licensed operator's qualification domain — consequence assessment, system state evaluation, regulatory accountability — is what makes them the appropriate Command Broker for above-Bright-Line actions in this environment. Delegation to a non-licensed person is not permitted for above-Bright-Line confirmations. The licensed-operator credential is necessary and not sufficient: the Broker is a role qualified by competence to judge the actions its Line places above the threshold, and that competence is not supplied by another credential (*The Broker* §4.1, §4.3), so the licensed operator must additionally hold the Command Broker qualification for the action classes concerned. Where an action class is discharged in advance under §3.1, the envelope author is the Command Broker discharging in advance, and the qualification required is the advance-mode competence within the Command Broker qualification, distinct from the in-the-moment competence and held within the one role (FD-BL-D1 §7; CB-IB_Qualification_Standard §4.1a).

Where action classes in the CntlCo environment are discharged in advance, configuration state is a moving baseline. An outage, a temporary modification to protection or telemetry, or a change in operating configuration entered after composition can put the system outside an envelope without anyone touching the envelope. The product carries the envelope status required at §3.8, so a suspended envelope cannot be invoked. Beyond that, the asset owner's change screening should trace each configuration change outward and flag every envelope it reaches at the moment the change is entered (*Pre-Approved Action Envelope* §5.4). This is stated as recommended practice rather than as a requirement because the method for mapping envelope dependencies onto configuration items is not yet developed (PAE-OI-9). Until it is, the flag rests

on the screening the facility already performs, and fails toward suspension where the dependency is uncertain.

DEFAULT (Must): The EMS Gen AI advisory product ships with a consequence classifier that routes each recommended action to either a below-Bright-Line confirmation pathway or an above-Bright-Line Command Broker confirmation pathway, and the interface is built to the full CB-UC-1 §4 requirement set. (a) Below-Bright-Line actions require a deliberate two-step select-then-confirm gesture (R-BL-2), with an audit record generated at confirmation (R-BL-4). (b) Above-Bright-Line actions require licensed system operator credential authentication, evidence of Command Broker qualification for the affected action classes, and the full CB disclosure set in the option panel — consequence characterization, anomaly flags, confidence-and-basis summary, and the action boundary statement (R-AL-1); the action boundary statement is retained, not displaced by the credential requirement, because it is what keeps an authorization from being read as authorizing a class of similar actions rather than this action, at this magnitude, under this system state. (c) The above-Bright-Line confirmation enforces a deliberate review pause — a minimum display time of the disclosure fields, not less than five seconds and configurable upward by the AOO — before the confirmation input becomes available, and that confirmation input is physically and visually separated from the selection input and labeled with language referencing the Command Broker obligation (R-AL-2), and the pause resets whenever a material change in conditions alters the consequence characterization or anomaly status of the pending option (R-MIX-3). (d) Options whose underlying system state has gone stale, or whose state diverges from the independent confirming channel beyond the stated tolerance, are visually flagged and their confirmation input disabled; for above-Bright-Line options the input is re-enabled by confirmation against the independent path, not by acknowledgment of the flag (R-GEN-4, R-MIX-4; FD-BL-D2 I3). (e) An emergency override is provided: under an identified emergency the operator may suspend the review pause through a distinct, explicitly labeled override input that generates a distinct audit flag; the override compresses the confirmation but does not reduce the operator's Command Broker accountability (R-AL-3). A product built without this pathway forces the wrong resolution to a fast-developing frequency excursion — moving actions below the line — which CB-UC-1 §3.3 explicitly declines to do. (f) Operator independence is preserved (R-IND-1 through R-IND-3), instantiating the cross-sector requirement in Part I §3.4: the advisory display does not obscure the independent system-state monitoring the operator needs; dismissing a recommendation is consequence-free and is not an audit event, while confirming one is; and Gen AI recommendations default to a lower urgency treatment than primary system alarms. (g) Execution of any confirmed action is contingent on the successful write of its audit record (R-AL-4).

DOWNGRADE PATHWAY (Should): Placement is not configurable. The consequence classifier may not be configured so that actions the DBA-ES-CntlCo instrument places above the Bright Line route to the below-Bright-Line confirmation pathway; that is defeat of the enforcement mechanism by configuration change, which FD-BL §4.5 forecloses. Where an AOO nonetheless requires it, the vendor must require a positive documented configuration action

identifying the specific action categories affected, the analytical basis advanced for the reclassification, and the compensating oversight architecture in place, and must state that the configuration record transfers the compliance obligation rather than altering the placement.

RISK TRANSFER OBLIGATION: When above-Bright-Line actions — as defined in DBA-ES-CntlCo — are confirmable without licensed operator credential authentication and Command Broker qualification for the affected action classes, documentation must state: (a) that the Command Broker qualification requirement under DBA-ES-CntlCo §7.1.2 is not satisfied for the affected action categories; (b) the specific actions and the reclassification basis; and (c) the AOO's acknowledgment and date of authorization.

8.4 Inverter Firmware Trustworthiness — DBA-ES-GN-FC8

DBA-ES-GN-FC8 establishes inverter firmware trustworthiness as a precondition of the determinism carve-out. An inverter operating on unverified firmware cannot rely on that carve-out (FD-LD §4.3), because the stated input bound over which its behavior is deterministic cannot be shown to hold as deployed, regardless of apparent normal behavior. Independent firmware verification is therefore a precondition for treating an inverter-based resource (IBR) as outside Bright Line analysis. This applies to approximately 80–85% of deployed U.S. utility inverters by manufacturer origin.

For inverter vendors and IBR control system vendors, the product requirement follows directly: the vendor must provide a documented firmware verification pathway that allows the AOO or an independent assessor to confirm firmware integrity without relying solely on the vendor's supply chain assurance. The three-question test in §3.3 applies to the inverter control firmware: it is deterministic within a stated input bound and formally testable as deployed only if the firmware has not been compromised. Trustworthiness — supply chain verified, not merely assumed — is the precondition for the determinism claim, and where the claim fails the firmware is an in-scope component whose actions are then placed on their own consequence, action by action.

DEFAULT (Must): The inverter or IBR control system product ships with a documented firmware verification pathway. The pathway must allow the AOO or an independent third-party assessor to confirm firmware integrity through a mechanism independent of the vendor's own supply chain assurance — including cryptographic hash verification against a vendor-maintained, externally auditable manifest. Firmware versions that cannot be independently verified do not satisfy the carve-out precondition under DBA-ES-GN-FC8, and the component is treated as in scope until they do.

DOWNGRADE PATHWAY (Should): Where an AOO operates inverters that lack an independent verification pathway — including legacy deployed units — the vendor should provide a compensating control specification: the minimum monitoring and anomaly detection architecture that must be in place before the inverter may be operated as an interim posture, with its above-line actions routed to authorization under §3.1. A compensating control package is not

equivalent to verified firmware; it is a documented interim posture pending verification pathway availability.

RISK TRANSFER OBLIGATION: When an inverter or IBR control product is deployed without a completed independent firmware verification, documentation must state: (a) that the firmware trustworthiness precondition under DBA-ES-GN-FC8 has not been satisfied; (b) that the inverter is an in-scope component pending verification, so that each control action it can take is placed on its own consequence and those above the line require authorization under §3.1; and (c) the AOO's acknowledgment and date of authorization.

8.5 Natural Gas Processing — Optimization-Safety Boundary

DBA-GAS-FC2 establishes the optimization-safety boundary as a formal Design Basis requirement — a series first. In continuous-process gas processing facilities, the Bright Line must constrain the plant's own commercial optimization logic, not only adversarial manipulation. Advanced Process Control (APC) and Real-Time Optimization (RTO) regions must be established as pre-authorized envelopes in the FD-EV §4.1 sense, including safety margin setbacks from SIS trip points authorized in advance by a qualified human and enforced by architecture. The same physical region, considered as a set of runtime limits to be monitored, is the operating band, and encroachment toward its operating limits is an event to detect and report (FD-EV §4.3).

For APC, RTO, and process optimization vendors serving natural gas processing facilities, this creates a product requirement that does not exist in traditional OT cybersecurity frameworks: the commercial optimization algorithm itself is a Bright Line candidate. If APC/RTO output can modify process parameters into the safety margin band without a safety instrumented system interlock or operator confirmation gate, the optimization system is operating above the Bright Line. The vendor must document where the APC/RTO operating limits end and the SIS trip point begins, and must confirm that the product architecture enforces the setback.

DEFAULT (Must): The APC/RTO product ships with a documented operating-limit specification that identifies, for each controlled process variable, the maximum setpoint the optimization algorithm may command without a safety interlock or operator confirmation gate. The boundary must be expressed as a measurable setback from the SIS trip point — not as a general design principle. The operating-limit specification, and the record of the human authorization establishing the pre-authorized envelope and its setback, are mandatory deliverables in the product's engineering documentation package.

DEFAULT (Must), continued: The product must additionally disclose the optimization objective function and its relationship to the safety margin, so that the person authorizing movement along the optimization gradient can see the objective the system is driving toward. This is the control against gradual margin erosion — the case in which each individual setpoint change is within bounds and correct on its own merits, but the aggregate trajectory walks the

process toward the safety margin. Trend visibility across authorizations, not per-action review, is what catches it.

DOWNGRADE PATHWAY (Should): A change to the setback itself is not a configuration option. Reducing or eliminating the safety margin setback moves the operating region toward a protective setpoint and is therefore a change to the pre-authorized envelope, categorically excluded from advance discharge under FD-BL-D1 R3 — no runtime gate can validate a change to its own safety basis. The product shall route such a change to in-the-moment Command Broker authorization by a qualified human, or to offline requalification producing a new authorized envelope and a new accepted completeness proof before it is fielded. Where an AOO nonetheless requires the change — for example, to maximize NGL recovery or throughput during peak demand — the vendor's guidance shall state that the requalification obligation applies and that a configuration record does not discharge it. The record shall identify the affected process variables, the modified setback margins, the compensating SIS or operator confirmation architecture, and the named human who authorized the new envelope and accepted its proof.

RISK TRANSFER OBLIGATION: When APC/RTO product output can reach SIS trip point proximity without a confirmed interlock or operator confirmation gate, documentation must state: (a) that the optimization-safety boundary requirement under DBA-GAS-FC2 is not satisfied by the default configuration, and that the change is an envelope change under FD-BL-D1 R3 requiring requalification rather than configuration; (b) the specific process variables and setback margins that have been modified; and (c) the AOO's acknowledgment, the compensating control architecture in place, and the date of authorization.

9. Nuclear Sector (DBA-NUC-SF)

9.1 Sector Overview

The nuclear sector, carried in the DBA series by DBA-NUC-SF (*Nuclear Sector Framework*), is the analytical origin of the Design Basis methodology that the entire DBA series extends to energy and other critical infrastructure sectors. Nuclear safety engineering formalized the DB→DBA progression in the 1960s and 1970s. The DBA-EN series inherits that methodology and extends it to deliberate military action. The nuclear sector's governance framework is therefore the most mature in the DBA series, and its product requirements carry the deepest regulatory grounding.

Nuclear facility I&C vendors operate in a regulatory environment defined by 10 CFR 50 Appendix B (quality assurance for nuclear power plant safety-related systems), IEEE 603 (criteria for safety systems in nuclear power generating stations), IEEE 7-4.3.2 (criteria for programmable digital devices in safety systems of nuclear power generating stations), and the NRC's cybersecurity rule under 10 CFR 73.54. The nuclear requirements of this chapter, and the

military-action extension DBA-MA applies to them, do not replace these regulatory requirements — they extend them to address military action threat scenarios that existing regulatory frameworks did not anticipate. References in this chapter to IEEE 7-4.3.2 are to the 2016 edition, which NRC Regulatory Guide 1.152 Revision 4 endorses. IEEE has approved a 2026 edition, which supersedes the 2016 edition at IEEE but has not yet been endorsed by the NRC. A vendor qualifies to the edition the licensee's licensing basis names, and should expect the 2026 edition to become the reference once the NRC endorses it.

The Bright Line in the nuclear context is the Bright Line as FD-BL defines it: a threshold on actions, placed on consequence, action by action. The nuclear regulatory framework does, however, supply an unusually well-developed architectural home for enforcement — the isolation and independence requirements applicable to safety systems, under which a failure in a non-safety system cannot prevent safety system action or cause spurious actuation. That isolation is where interposition is built in a nuclear plant. It is not where the line falls, and a component's position on the non-safety network neither places its actions below the line nor removes them from analysis. This chapter applies the placement criterion to in-scope components (FD-LD §4.3): each action is placed on its own consequence, and what an incorrect change to the parameter could cause, not the safety classification of the system that carries it, is what puts the action above the line (CB-Annex-Nuclear §3.2).

9.2 I&C Vendor Landscape

Nuclear I&C vendors fall into two primary categories for the purposes of this chapter. The first is the safety system I&C supplier — vendors of reactor protection systems, engineered safety feature actuation systems, and safety-related control systems. These vendors operate under 10 CFR 50 Appendix B quality assurance requirements and must demonstrate software qualification under IEEE 7-4.3.2. The second is the cybersecurity overlay vendor — vendors of OT security monitoring, network segmentation, anomaly detection, and incident response tools deployed in the nuclear OT environment. These vendors must operate within the 10 CFR 73.54 cybersecurity rule framework.

Gen AI tools are emerging in both categories: AI-assisted safety system diagnostics, Gen AI operator advisory tools for emergency procedure guidance, and ML-based anomaly detection for nuclear OT networks. The requirements of this chapter address each category. The Bright Line applies with particular rigor in the nuclear context because the consequences placed above it are the most severe in any sector, and because the regulatory framework already supplies the architecture in which interposition can be enforced.

9.3 Placement in the Nuclear Context and the Safety/Non-Safety Isolation Architecture

Placement in the nuclear context follows CB-Annex-Nuclear §3.2: a change initiated or materially influenced by an in-scope component falls above the Bright Line where the consequence of that change places it there. Material influence decides whether the obligation reaches a change already placed above the line on its consequence, and never whether the change is above the line (FD-BL-D1 §7). The parameters this reaches are identified by what an incorrect adjustment could cause, and not by the safety classification the supporting system carries. They include changes bearing on reactivity, on credited heat-removal flow paths, on spent fuel pool makeup and cooling, on safety electrical bus alignment, and on containment isolation or integrity. The safety/non-safety isolation established by the applicable safety-system criteria is the architecture through which that placement is enforced, not the criterion by which it is made. The principle of that architecture is that safety systems must be isolated from non-safety systems such that a failure in a non-safety system cannot prevent safety system action or cause spurious safety system actuation.

[Editorial note, this version: IEEE 7-4.3.2 was confirmed against current editions at v0.11 (see §9.1); the other safety-system standard and clause references supporting this section are to be confirmed before issue, consistent with FD-BL §8. This chapter also issues pending Project 6 concurrence.]

For AI and Gen AI components, this means:

- An in-scope advisory component (FD-LD §4.3) operating on the non-safety (balance of plant) network must have no electronic pathway to a safety system parameter without human interposition. The licensed Senior Reactor Operator is the natural locus of the Command Broker function for reactor process parameters, and the requirement maps onto the existing licensed-operator structure rather than displacing it. The licensed-operator credential is not by itself a Command Broker qualification: the Broker is a role qualified by competence to judge the actions its Line places above the threshold, and that competence is not supplied by another credential (*The Broker* §4.1, §4.3). What the licensed operator must add to satisfy the Command Broker requirement is specified in CB-Annex-Nuclear and the CB-IB Qualification Standard, and the vendor's obligation is to supply the artifacts those instruments require (§3.5).
- An ML component operating within a safety system must be qualified under IEEE 7-4.3.2 as a software component of a safety-related system. The determinism and formal testability property test that governs scope (FD-LD §4.3) aligns with, and for safety-related software is subsumed by, the software qualification requirements of the nuclear regulatory framework.
- A cybersecurity monitoring tool operating on the non-safety network that generates alerts or recommendations related to safety system status must not have a direct pathway to safety system actuation or parameter change. The alert's downstream action — if any — is placed on its own consequence, and where it falls above the line it must pass through a qualified Command Broker.

DEFAULT (Must): I&C products intended for nuclear applications enforce the safety/non-safety isolation as an architectural property, and that isolation is the architecture through which above-line placement is enforced. No component in scope under FD-LD §4.3 has a pathway, direct or through integration, to a safety system parameter without interposition by a qualified Command Broker.

DOWNGRADE PATHWAY (Should): Where an integrator or AOO proposes a deployment configuration that creates a pathway from a component in scope under FD-LD §4.3 to a safety system parameter without interposition by a qualified Command Broker, the vendor must provide written guidance stating that such a configuration defeats the interposition required for above-line actions in this sector and that any regulatory deviation must be evaluated under the 10 CFR 50.59 change control process.

RISK TRANSFER OBLIGATION: Vendor representations that a Gen AI or AI-assisted tool is suitable for safety-related nuclear applications must be supported by a software qualification pedigree under IEEE 7-4.3.2 or an explicit statement that the tool is classified as a non-safety system and is subject to the isolation requirement above.

9.4 Safety System Software Qualification

Safety-related software in nuclear applications must be qualified under IEEE 7-4.3.2. The qualification process includes: (1) software development lifecycle documentation; (2) formal verification and validation (V&V), including software hazard analysis; (3) independence requirements for V&V activities; and (4) configuration management throughout the software lifecycle. These requirements apply to any ML component integrated into a safety-related system.

The three-question test in §3.3 applies with a nuclear-specific fourth question for safety-related applications: Has the ML component been qualified under IEEE 7-4.3.2 or the applicable NRC-accepted alternative? The answer must be yes for any ML component embedded in a safety-related system. A component whose outputs are not reproducible under identical inputs cannot be qualified under the existing nuclear software qualification framework, whatever its architecture is named, and therefore cannot be integrated into a safety-related system under current regulatory requirements.

DEFAULT (Must): Vendors of safety-related I&C systems document the software qualification pedigree of all embedded ML components, including the applicable IEEE standard under which qualification was performed, the independent V&V organization, and the configuration management record.

DEFAULT (Must): Vendors of safety-related I&C systems do not integrate non-deterministic components into safety-related software. Where a product includes both safety-related and non-safety functions, those components are architecturally isolated from the safety-related software in a manner verifiable under 10 CFR 50 Appendix B.

DOWNGRADE PATHWAY: No downgrade pathway exists for the exclusion of non-deterministic components from safety-related systems under the current regulatory framework. Any proposed integration of non-deterministic AI into a safety-related system requires an NRC exemption or rulemaking.

9.5 Cybersecurity Overlay Vendors — Nuclear OT

Vendors of cybersecurity tools deployed in the nuclear OT environment operate under the 10 CFR 73.54 cybersecurity rule. AI-based anomaly detection tools deployed in the nuclear OT environment carry the scope determination at §2.1. An ML-based anomaly detection tool that is deterministic within a stated input bound and formally testable as deployed is outside the Command Broker obligation under the carve-out — which is not a finding that its actions are below the line (FD-LD §4.3). However, any response action triggered by the tool that could affect safety system function, security system function, or emergency preparedness system function requires operator interposition before execution.

DEFAULT (Must): Nuclear OT security monitoring tools ship with automated response actions that affect safety, security, or emergency preparedness functions disabled by default. Automated response actions in those consequence categories are enabled only by a positive AOO configuration action that transfers the compliance obligation rather than altering placement, with documented authorization, auditable log entry, and risk transfer documentation consistent with §3.1. These automated response actions, including isolating a segment or shutting down a service, are actions under FD-BL §4.3, and the product places each through the consequence classifier of §3.2.

DOWNGRADE PATHWAY (Should): Where an AOO enables automated response actions affecting safety-adjacent systems, the vendor must require a documented configuration action reviewed by the AOO's security organization and consistent with the facility's 10 CFR 73.54 cybersecurity plan.

9.6 Red-Teaming in the Nuclear Context

Red-teaming of Gen AI components in nuclear non-safety applications follows the requirements in §3.6, with the addition of a nuclear-specific attack scenario: the adversary's goal in a DBA-MA nuclear scenario is not simply to compromise a system but to cause a safety function to fail or to trigger spurious safety actuation while evading detection. Gen AI tools deployed in the nuclear environment must be red-teamed against scenarios in which the adversary has crafted inputs designed to produce recommendations that would lead an operator to take an action inconsistent with safety function requirements.

This is the nuclear sector analog to the adversarial injection requirement in §8.3.1: the adversary is not attacking the system directly but is attacking the information presented to the human

Command Broker — the licensed SRO — to induce an unsafe decision. Red-team scope for nuclear Gen AI advisory tools must explicitly include this attack vector.

10. Emerging Sectors

The following sectors have active DBA development work in progress. Substantive vendor requirements for these sectors will be incorporated into future DBA-VC revisions as the applicable sector frameworks mature. Vendors serving these sectors should consult the applicable DBA framework documents directly and engage with Eclectic Technologies for current guidance.

10.1 Water Sector (DBA-W)

The DBA-W Sector Framework is the most mature non-energy, non-nuclear DBA instrument. §5.5 of DBA-W (Water Sector Framework), as published at v1.16 in CIKR Design Basis Examples, Edition 1 (November 2026), carries the most fully developed sector-specific implementation of the Command Broker and Bright Line requirements currently in the series. Vendors serving water sector clients should read DBA-W §5.5 as the current authoritative source for water-sector AI governance product requirements. A substantive water-sector chapter will be incorporated into DBA-VC on the next scheduled revision.

Vendors who built to §5.5 at an earlier version should note that its tier statements changed materially at v1.12. An earlier form permitted autonomous operation at the highest criticality tier on the ground that a component's behavior was bounded and testable; that ground is withdrawn. Determinism within a stated input bound, formally testable as deployed, is a carve-out from the scope of the Command Broker obligation under FD-LD §4.3, and is not a finding that the actions concerned fall below the Bright Line. Part I of this document states the same rule and governs here.

10.2 Oil and Gas Sector

The DBA-GAS and DBA-OIL sector frameworks carry Command Broker and Bright Line requirements in gas and petroleum sector operational vocabulary (§5.4 of each). The optimization-safety boundary requirement established in DBA-GAS-FC2 (see §8.5 of this document) is the most distinctive energy-sector-specific requirement for APC and RTO vendors. The sector applicability matrix in §4 provides the current reference mapping. A dedicated oil and gas sector chapter will be incorporated in a future revision.

10.3 Transportation and Other Critical Infrastructure Sectors

DBA series development for transportation, communications, and other NIPP critical infrastructure sectors is in the planning stage. Vendors serving these sectors who wish to engage with the DBA series development process should contact Eclectic Technologies directly.

References

DBA-EN series instruments are cited in this document by designation, without version tokens. The current version of each is stated in DBA-EN-REG, which is the authoritative register for this series; a version-bearing citation would duplicate that register and drift from it between revisions. Resolve any series instrument named in this document against DBA-EN-REG. External regulatory references and supporting documents retain their version and date tokens.

Command Broker family instruments are cited in this document by designation, without version tokens. The current version of each is stated in CB-REG, which is the authoritative register for this family.

Foundational Definitions and Determinations

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Roxey, Timothy E. Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector). Eclectic Technologies, 2026.

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Revision History

Ver.	Date	Author	Description
0.1	Jun 2026	T. Roxey	Initial working draft. Established cross-series vendor companion document type. Six canonical product requirements derived from DBA-W §5.5, DBA-GAS §5.4, and DBA-DC §3.5. Default/downgrade/risk-transfer obligation structure introduced. Sector applicability matrix and highest-criticality tier table completed. The procurement signal cascade argument has been developed. Evidentiary package elements identified.
0.2	Jun 2026	T. Roxey	Added §3.7: EMS CntlCo-specific requirements. Three interaction points from DBA-ES-CntlCo v0.2 §7 translated into product specifications: state estimator ML advisory-only boundary; Gen AI operator decision support with licensed operator as mandatory Command Broker; adversarial input robustness as a distinct product security requirement. Sector applicability matrix updated. §2.3 Bright Line note revised to reflect correction of “Federal Coordination Threshold” usage in DBA-GAS and DBA-OIL.
0.3	Jun 2026	T.	Major restructuring. Reorganized into Part I (Cross-Sector Foundation) and Part II (Sector

		Roxey	Chapters). §2.4 Cross-Sector ICS Amplifier added to Part I as a core framing argument. §2.5 Vendor Applicability Self-Assessment added with narrative guidance and one-page checklist. Part II §8 (Energy Sector) established, absorbing prior §3.7 CntlCo content into §8.3 and adding §8.4 (Inverter Firmware Trustworthiness) and §8.5 (Optimization-Safety Boundary). Part II §9 (Nuclear Sector) added as substantive chapter. Part II §10 (Emerging Sectors) added as framework stubs. Sector applicability matrix updated. References expanded to include the nuclear regulatory framework.
0.4	Jun 2026	T. Roxey	Energy chapter expanded. §8.4 (Inverter Firmware Trustworthiness) and §8.5 (Natural Gas Processing — Optimization-Safety Boundary) brought to full product specification standard: Default/Downgrade/Risk Transfer structure added to each. §8.1 and §8.2 version references updated (DBA-EN-SF v0.7→v0.8; DBA-ES-TX-FC2 v0.2→v0.3; DBA-ES-GN-FC8 v0.1→v0.2). References updated to match. Gap N-19 closed (CntlCo EMS vendor specification confirmed complete in v0.3 §8.3). Superseded by v0.5 due to errors in §8.3.1 and §8.3.2; deleted.
0.5	Jun 2026	T. Roxey	§8.3.1 and §8.3.2 revised to correctly reflect V29 §5.3 and CB-Framework Bright Line architecture. §8.3.1: Removed incorrect advisory-only restriction on ML. Established QA obligations (random error tolerance + adversarial injection tolerance) as the governing condition for closed-loop autonomous ML operation. Introduced Gen AI/ML layered architecture and consequence classifier as the Bright Line enforcement mechanism. Absorbed adversarial injection tolerance requirement (formerly §8.3.3) into §8.3.1 DEFAULT block. §8.3.2: Removed categorical “Gen AI tools operate above the Bright Line” statement. Established action-specific Bright Line placement per CB-UC-1. Rewrote DEFAULT block around consequence classifier and two-pathway confirmation architecture. Retained licensed operator as mandatory Command Broker for above-Bright-Line confirmations. §8.3.3 removed (content absorbed into §8.3.1).
0.6	Jul 2026	T. Roxey	Absorbed CB-UC-1, the CB-IB Qualification Standard, and the Command Broker Implementation Guide as sources. §2.3 reconciled to the consequence-threshold model with the application-level (tier) vs. action-level (Bright Line placement) distinction stated explicitly; §2.2 tier crosswalk added. New canonical §3.7 (audit record requirements, from CB-UC-1 §5). §3.1 conformance-and-misrepresentation clause added. §3.5 extended with three vendor-supplied artifacts (operational profile summary, proactive update notification, qualification exercise mode); §6.1 evidentiary package updated to match. §8.3.2 DEFAULT rebuilt against the full CB-UC-1 §4 R-AL/R-MIX/R-IND set (action boundary statement restored; review pause, staleness flagging, emergency override, operator independence, and audit-write contingency added). §8.5 optimization-objective disclosure added. Source study citation corrected to the canonical Eclectic Technologies edition. Mechanical fixes: §9.6 reference §8.3.3→§8.3.1; §2.5 water-sector reference §3.1→§10.1; ‘Fourth Edition’ removed from cover. Traceability table added.
0.7	Jul 2026	T. Roxey	Reconciled to the Consequence-Severity Tier Model. §2.2 recast as criticality bands on the six-category take-the-max basis; naming crosswalk demoted to legacy and the type-words returned to the error

			classes; §2.3 band/action distinction clarified against the error-class axis; obligation structure relabeled ‘obligation levels’ to free ‘tier’ for criticality; band vocabulary applied throughout.
0.8	Jul 2026	T. Roxey	Restored the cross-sector operator-independence requirement to Part I §3.4 (recovered from the June v0.6 draft; §8.3.2 now references it as the energy instantiation). Restated the governs-which-instrument clause in §3.1 (a vendor conforms to DBA-VC, not the Implementation Guide; inputs supplied per §3.5) — the §2.3.1 Convergent Origination framing stays retired. Ran the version-citation pass: CB-Framework→v0.2, Implementation Guide→V0.2, CB-CT→v0.2.
0.9	Aug 2026	T. Roxey	Full conformance revision against the FD governing stack, applying DBA-VC v0.8 Conformance Markup v0.1. GOVERNING CITATIONS. Governing-definition block added at the head of Part I citing FD-BL v0.3, FD-BR v0.1, FD-LD v0.1, FD-BL-D1 v0.3, FD-BL-D2 v0.2 and FD-EV v0.1; References gained a Foundational Definitions and Determinations block; the document propagates by reference rather than restatement (FD-BL §2.1). SCOPE VS PLACEMENT. §2.1 rebuilt around the FD-LD §4.3 property test and the stated-input-bound qualifier, and its closing paragraph conformed; §2.5 step three, §8.3.1 ¶2, §8.4 and §9.5 corrected — determinism is scope, not placement, and the carve-out is not the line (FD-BL §4.4). §3.3’s three questions restated on the property test and on action-level placement. §3.6 red-teaming rescoped from technology class and band to in-scope components producing above-line actions. GRANULARITY. §2.2 band descriptions recast as statements of assessment rigor; band-level placement struck from all three bands (FD-BL §4.3). §8.4 recast from system-level to action-level placement. MODE OF DISCHARGE. §3.1 restated around the two discharge modes and the two-object model (FD-BL-D1 R1–R3, §5); §2.3 ¶2–3 conformed to match — this exceeds the markup, which authorized only the citation move, and the matching amendment is owed to CB-Framework §3.1/§3.2 which still states the human-broker requirement categorically. §8.3.1 ¶3 composite-classification statement struck and the layered architecture restated as mode of discharge under R1–R3 (FD-BL-D1 §6). §8.5 setback change routed to requalification under R3. ENFORCEMENT. §3.2 rebuilt around the consequence classifier in place of a single gate at the advisory/command seam, and retitled. Downgrade pathways at §3.1, §3.2, §8.3.2 and §9.5 conformed to FD-BL §4.5 — the enforcement

			mechanism is not defeasible by configuration, and placement is not configurable; what a downgrade transfers is the compliance obligation. INDICATION INTEGRITY. §3.4 gained an indication-integrity DEFAULT (FD-BL-D2 I1–I3, denied-indication behavior), was rescoped off the Gen AI label and off the criticality band, and was retitled; §8.3.1 DEFAULT gained the corroboration path; §8.3.2(d) staleness treatment raised from acknowledgment to corroboration; §8.3.2(c) restored the R-AL-2 separated, labeled confirmation input. NUCLEAR. §9.1 and §9.2 closing paragraphs rewritten and §9.3 retitled and rewritten: placement conformed to CB-Annex-Nuclear v0.1 (action-level, on consequence), the network-boundary framing withdrawn from the body, the §4 matrix and the §9.3 DEFAULT, and the isolation architecture retained as the means of enforcement rather than the criterion. §9.3 and §8.3.2 note that the licensed-operator credential is not by itself a Command Broker qualification (FD-BR §4.3). Clause-level standard citations (IEEE 603 §5.1, 10 CFR 50 App. A GDC-24) withdrawn pending verification against current editions (FD-BL §8); the document-level IEEE 603 reference is retained. §9 issues pending Project 6 concurrence. TERMINOLOGY. FD-EV reservation notice added at §2.3; nine ‘envelope’ uses conformed — §8.5’s APC/RTO region named a pre-authorized envelope for its authorization role and an operating band for its monitored limits, §8.3.1 and §8.3.2 renamed to operating band and operating limit. CROSS-REFERENCES AND HOUSEKEEPING. §4 matrix gained the §3.7 row; §3.2 traceability re-anchored to FD-BL §4.2–§4.3 and CB-Framework §3.1/§3.2a, §3.1 and §3.4 rows gained their governing instruments, §3.3 row narrowed to CB-IB §4.1; §3.1’s statement of the Implementation Guide’s standing corrected against that Guide’s own §1 (CB-IB governs competency); §3.7 audit IDs reordered to 1-2-3-4, the R-AUD-1 documented-schema hardening recorded as deliberate, and the record extended to advance-discharge authorizations; §6.1 gained the gatekeeper acceptance-set proof and the indication-integrity design statement, and its component-classification artifact was conformed to action-level placement. CITATIONS REFRESHED. DBA-EN-SF v0.10, DBA-ES-CntlCo v0.4, DBA-ES-TX-FC3 v0.3, DBA-ES-GC-FC4 v0.5, DBA-MA Introduction v1.10, DBA-W v1.7, DBA-DC-L1 v1.9, FD-BL v0.3; CB-Annex-Nuclear v0.1 added to References. AUTHOR DECISIONS TAKEN. §8.3.2’s battery-dispatch region read as an operating band, not a pre-authorized

			envelope, there being no advance-authorization apparatus stated for it (FD-EV §4.1). The R-AUD-1 schema hardening retained deliberately. Part I §3 not restructured; the requirement is instead stated once at the head of Part I and applied thereafter. FORMATTING. Nine paragraphs bolded end-to-end in v0.8 restored to the house run-in-label pattern; footer version corrected. The scope-wording pass (in-scope component, not the technology label) was carried through §2, §3, §5.3, §6.1, §8.4, §9.3, §9.4 and §9.5. Also conformed outside the markup's edit list: §3.3's three questions and DEFAULT, the §2.5 checklist scope row, §6.1's component-classification artifact, §9.5's opening cross-reference, and the downgrade definition at the head of §3. Harvest items recorded in FDBL Upward-Harvest Register v0.1. Outstanding coordination: DBA-GAS-FC1 (cover reads v1.5, filename reads v1_6) and DBA-OIL-FC1 (cover reads v1.3, revision history ends at 1.2) are cited here from their filenames and should be reconciled at source; CB-Framework §3.2 (categorical human-broker statement) and §3.3 (Mid and Lowest band descriptions) are now behind this document and require the matching amendment.
0.10	Sep 2026	T. Roxey	Citation-form conformance pass, the first run on this instrument; no rule was added, removed or changed and no requirement moved. Forty-six version tokens were enumerated and forty-four dispositioned. STRIPPED, resolving to their registers: the FD governing stack (FD-BL, FD-BR, FD-LD, FD-BL-D1, FD-BL-D2, FD-EV), which this project pins nowhere; the Command Broker family members CB-REG records (CB-Framework_Cross_Sector, CB-IB_Qualification_Standard, CB-UC-1, CB-CT); and the in-series DBA-EN citations (DBA-EN-SF, DBA-ES-CntlCo, DBA-ES-TX-FC2, DBA-ES-TX-FC3, DBA-ES-GC-FC4, DBA-ES-GN-FC8, DBA-ES-MA-SF, DBA-GAS-FC1, DBA-GAS-FC2, DBA-OIL-FC1), resolving to DBA-EN-SF §1.4. REPAIRED rather than stripped, being instruments this project does not own or that no register resolves: DBA-W v1.7 to v1.12, DBA-MA Introduction v1.10 to v1.11, CB-Annex-Nuclear v0.1 to v0.2, and DBA-DC-L1 v1.9 to v1.11 — the last because §1.4 records Critical Loads as a single cell and does not carry the member. LEFT PINNED: Command Broker Implementation Guide V0.2, which CB-REG does not record. Both canonical lead-ins added to References. §10.1 gains a currency note: the v1.7 text it directed vendors to carries the pre-R3 formulation withdrawn at DBA-W v1.12, raised by Project 9 on 8 September 2026. The DBA-EN-SF pin stood at v0.10 against a v0.17

			master. Revision-history entries were excluded from the pass and read afterwards to confirm the exclusion held.
0.11	Oct 2026	T. Roxey	Citation-condition revision, on the author's ruling of 28 September 2026 that DBA-VC is Project 8's and cites the Command Broker family's instruments rather than restating them. Part I §3.1's in-the-moment confirmation now cites CB-Framework_Cross_Sector §3.1 for its properties; §3.4's operator-independence paragraph cites CB-UC-1 R-IND-1 to R-IND-3, applied with each sector's own parameters under CB-UC-1 §7; §3.7 cites CB-UC-1 §5 and R-AL-4, keeping its two additions (the documented schema made mandatory, and the FD-BL-D1 §8 record for advance discharge); §3.5 names the family instruments that rely on items (e) to (g). No requirement is weakened. IEEE 7-4.3.2: the 2016 edition, endorsed by RG 1.152 Rev. 4, stays the reference, and the 2026 edition, approved by IEEE and not yet endorsed by the NRC, is noted at §9.1 and added to References; the §9.3 editorial note is answered for this standard, and RG 1.152 is corrected from Rev. 3 to Rev. 4. §1 no longer says the water framework has been published; it has not. The series lead-in names DBA-EN-REG instead of DBA-EN-SF §1.4; CntlCo's new title is carried; version tokens are stripped from seven Command Broker family and DBA-EN citations that resolve to CB-REG or DBA-EN-REG; CB-Annex-Nuclear's stale v0.2 pin was among them.
0.12	Oct 2026	T. Roxey	Conformed to the October 2026 Edition of the Foundational Definitions, on Project 0's work order of 3 October 2026 (drift-screen finding P8-03 and the notes of 2 October), and to the Command Broker family as CB-REG records it after Project 2's conformance of the same day. NUCLEAR. §9.3's placement sentence follows CB-Annex-Nuclear §3.2 as that annex now reads: a change initiated or materially influenced by an in-scope component falls above the line where its consequence places it there, and material influence is a test of reach, never of placement (FD-BL-D1 §7); the parameters are identified by what an incorrect adjustment could cause, not by safety classification. §9.1's closing sentence and §9.3's balance-of-plant sentence are conformed to match, the latter stating its reach by scope (FD-LD §4.3) rather than by technology, and the stale CB-Annex-Nuclear pin in the §4 matrix, missed at v0.11, is stripped. MODE OF DISCHARGE. The verb is aligned with FD-BR §4.4 and FD-BL-D1: the Command Broker discharges in advance by accepting the envelope and its proof, and the verified gatekeeper admits. §3.1's advance-discharge mode, §8.3.1's layered architecture, §3.1's

			governs-which-instrument clause, §3.7, §6.1 and §8.3.2's envelope-author qualification now read that way, the last citing the advance-mode competence held within the one role (CB-IB_Qualification_Standard §4.1a). §3.1 adds that an envelope with no verified gatekeeper admitting against it is not an authorization (CB-PAE). CLASSIFIER INPUTS. §3.2 gains a DEFAULT on the consequence classifier's inputs (FD-BL-D2 §4.1 to §4.3): traceable to independently verifiable values, and, where uncorroborated, the affected action does not proceed without in-the-moment Command Broker authorization, a rule of discharge that changes no placement; the §3.2 traceability row cites it. TERMINOLOGY. "Which instrument governs" and the governing-definitions block drop the retired FD-BR subtitle and say that a verified gatekeeper may stand in for the Broker's presence. CITATIONS. The carve-out is cited to FD-LD §4.3 throughout in place of FD-BL §4.4, including the CntlCo and FC8 paragraphs; §8.3.1's state-estimator sentence states the carve-out as a conditional property test rather than a class assertion and as no grant of ungoverned autonomy. The cover date, which still read August 2026, is corrected. Revision-history entries were excluded from every substitution and read afterwards to confirm the exclusion held. Publication handling under the CR-PB draft: the published v0.8 is already diverged, so this pass acquires no republish step.
0.13	Nov 2026	T. Roxey	Conformed to the November 2026 Edition of the Foundational Definitions, applying the conformance markup redone on 9 November 2026 on the author's approval of the same day. The carve-out holds only for the configuration formally tested (FD-LD §4.3), and scope reaches every component not meeting the carve-out rather than non-deterministic components only, at §2.1 and §2.5. §3.5 gains a DEFAULT under which a vendor release is taken in only through a governed act, staged releases are placed on their whole planned reach, and a release ends the carve-out until the new configuration is tested; §6.1 records the governed act. On the author's ruling of 9 November 2026, the §8.3.1 downgrade pathway no longer authorizes closed-loop autonomous operation on a compensating record: a component without a complete QA package is in scope and its above-line actions route to the Command Broker, and the §8.3.1 references to closed-loop autonomous operation are conformed to operation under the carve-out. §3.2 names the kinds of change FD-BL §4.3 counts as actions, enterprise-network changes among them, and §9.5 places

			nuclear OT automated responses through that classifier. §3.4 takes FD-BL-D2's sufficiency paragraph. No requirement is withdrawn. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default, on the author's ruling of 9 November 2026.
0.14	Nov 2026	T. Roxey	Author review of 11 November 2026. §1 names the sector framework set as it now stands: a framework for each of the sixteen critical infrastructure sectors, energy carried by DBA-EN-SF and its three sub-sector frameworks with DBA-DC in the Critical Loads grouping, nuclear by DBA-NUC-SF, and DBA-MA named as the military-action extension rather than as the nuclear framework. The sentence stating which frameworks have been published is removed, because it carried state that drifts. §2 expands CB-IB at first use as Command Broker / Influence Broker, and its reference entry is expanded to match. FD-BR is given its title at first naming in §2 and in §3.1, and its section citations take the form The Broker §n at §3.1, §8.3.1, §8.3.2, §9.3 and the Part I governing-instrument table; the other Foundational Definitions are left in their existing form for now, on the author's instruction. §3.4 adds a plain-language reading of each of CB-UC-1 R-IND-1 to R-IND-3, CB-UC-1 continuing to state and govern the requirements. The §2.2 criticality-band text is reviewed and left as it stands. Nuclear references elsewhere in the document that treat DBA-MA as the nuclear framework (§2.4, the §4 matrix, the Part II §9 heading) are noted and not changed at this version. No requirement is added, withdrawn or changed. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.
0.15	Nov 2026	T. Roxey	Nuclear references conformed to the sector framework set, on the author's instruction of 11 November 2026, closing the item v0.14 noted. Where the document used DBA-MA to mean the nuclear sector framework it now names DBA-NUC-SF (Nuclear Sector Framework): §2.4's cross-sector transfer sentence, the nuclear column of the §4 sector applicability matrix, the Part II §9 heading, and §9.1's opening sentence. In §9.1 and §9.2 the phrases "DBA-MA nuclear sector requirements" and "for DBA-MA purposes" now refer to this chapter's requirements, the §9.1 sentence on extending regulatory requirements to military-action scenarios naming DBA-MA as the military-action extension, which it is; §9.1's placement sentence now reads that this chapter applies the placement criterion. Uses of DBA-MA in its own sense — the DB→DBA→DBA-MA

			progression at §8.1, the governing DBA-MA scenario at §8.3.1, and the DBA-MA nuclear scenario at §9.6 — are unchanged. References: DBA-NUC-SF is added at v0.7, and the DBA-MA Introduction citation moves from v1.11 to v1.14, both verified against the Project 6 masters on 11 November 2026. No requirement is added, withdrawn or changed. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.
0.16	Nov 2026	T. Roxey	The Command Broker interface thread is carried into the vendor companion, on the author's decisions of 11 November 2026. §3.4 opens by distinguishing the two interface archetypes through which the Command Broker works, the live console of CB-UC-1 and the deliberative composition interface, and states that the console's protections do not apply to composition. New §3.8 specifies the deliberative composition interface: a workspace in which the engineer and the broker assemble the picture from multiple analytical streams; the configuration baseline presented there rather than on the console; an interaction log as the composition record, the form the author chose; a reason for each bound selected from a controlled list that the asset owner's configuration management program owns, with brief free text where no reason fits or the bound departs from the proposal; indication integrity at composition; and envelope status carried on the envelope, restorable only through the configuration management program. New §3.9 specifies supplier model-change notification and supplies PAE-OI-5 in product-specification form: a material change is any change to the system nameplate. §3.5 gains the system nameplate, drawn from the nameplate discussion of the hybrid quality-assurance work; it identifies the deployed configuration rather than the model alone, may be issued in SPDX 3.0 or CycloneDX with the elements neither format carries as extension properties, and carries a weights digest only optionally, on the author's decision. §3.1 cites CB-PAE for the in-advance pathway; §3.7 binds the composition, approval and invocation records of each envelope; §6.1 adds the interaction log and the version binding to the evidentiary package and the nameplate to the provenance records; §8.3.2 states configuration-state flagging as recommended practice, PAE-OI-9 being unsolved. The Part I governing-instrument table and the §4 matrix gain rows for §3.8 and §3.9. References add CB-PAE, the hybrid formal methods paper and the two bill-of-materials standards. CB-PAE is cited while it remains an unpublished skeleton whose designation is still open (PAE-OI-1). PAE-OI-11 and PAE-OI-12, the

			nomination of envelopes from operational evidence, are held by the author and not carried. No existing requirement is withdrawn. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.
0.17	Nov 2026	T. Roxey	Equivalency acceptance added to §3.9, on the author's instruction of 11 November 2026. The facility's configuration management program may show, by a single acceptance evaluation, that a new nameplate is equivalent to the one it updates for the envelopes the evaluation names, and so carry those envelopes across together, on the pattern by which a nuclear plant accepts a replacement item that is not identical to the original. The equivalence is behavioral; the evidence is the vendor's per-element change description and regression results together with the asset owner's own acceptance testing; an envelope whose basis rests on a changed element is excluded and re-validated individually; and approval re-binds each covered envelope to the new identifier, recording both. On the author's decision, the prior configuration keeps running for envelope invocations until cutover, and the vendor keeps it available for that purpose; the DOWNGRADE pathway suspends the bound envelopes where the vendor cannot. §6.1 adds Equivalency Acceptance Support. CB-PAE does not yet provide for equivalency acceptance; the doctrine is proposed to Project 2 by note of the same day. No existing requirement is withdrawn. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.
0.18	Nov 2026	T. Roxey	Vendor-facing correction to §3.9, no requirement added, withdrawn or altered. The closing sentence of the equivalency acceptance provisions no longer names a project: it states that CB-PAE does not yet provide for equivalency acceptance, that this section carries it at product level until it does, and that the provisions will be conformed when CB-PAE states the rule. The run-in label and the §3.9 row of the Part I governing-instrument table mark the equivalency provisions provisional. Both changes follow from the author's ruling of 28 September 2026 that DBA-VC cites the Command Broker family rather than stating doctrine ahead of it. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.
0.19	Nov 2026	T. Roxey	Conformed to CB-PAE (Pre-Approved Action Envelope) at v0.7, on the author's decisions of 11 November 2026 and Project 2's return note of the same day. §3.9 cites Pre-Approved Action Envelope §5.2

			for equivalency acceptance, and the provisional marker is removed from the run-in label, from the §3.9 row of the Part I governing-instrument table, and by deletion of the closing sentence that stated it. The equivalency provisions take coverage stated envelope by envelope, with the elements each envelope's basis depends on and the evidence that the behavior it depends on is unchanged, so that a listing without reasons is a bulk restoration on which the product does not re-bind; an empirical evaluation whose acceptance criterion is stated before the test set is run; a test set drawn from the cognitive error taxonomy on which the tiered assessment method rests; and the statements that re-binding does not reset an envelope's expiry and that an accepted equivalency evaluation triggers the aggregate review. The §3.9 opening cites §5.2 for the assignment of the product-specification form, PAE-OI-5 being addressed. §3.8 adds that the reasoning belongs to the asset owner's configuration management record and enters the approval record's basis, and clause (b) of its risk-transfer obligation cites Pre-Approved Action Envelope §7.1. §6.1 Equivalency Acceptance Support and the §3.8 row of the Part I table are extended to match. No existing requirement is withdrawn. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.
0.20	Nov 2026	T. Roxey	Water-framework citation repaired, no requirement added, withdrawn or altered, on Project 10's note of 11 November 2026 and the author's decision of the same day. §10.1 and the DBA Series reference list had cited the water framework at v1.12, a version a vendor cannot obtain. Both now cite it as published at v1.16 in CIKR Design Basis Examples, Edition 1 (November 2026), with the edition's DOI; the reference entry follows the freeze copy's citation block. The sentence in §10.1 recording that the tier statements changed materially at v1.12 is a historical statement about that version and stands as written. Publication: the master is corrected and the publication is left standing, under the CR-PB interim default.